Palaeomagnetism and Sm–Nd ages of the Neoproterozoic diabase dykes in Laanila and Kautokeino, northern Fennoscandia

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Abstract: The results of palaeomagnetic, rock magnetic and isotope data on the NE–SW trending Neoproterozoic Laanila–Ristijärvi (Finland) and Kautokeino (Norway) dyke swarms, northern Fennoscandia, are reported. Positive baked contact tests in both dyke sets indicate that the characteristic remanence is primary thermoremanence. Rock magnetic and petrological properties show similarity between the two dyke swarms. All the dykes except one (where the dominant carrier is Ti-rich titanomagnetite) have mixed multidomain and single-domain or pseudosingle-domain magnetic carriers of Ti-poor magnetite. From its remanence directions, petrography and rock magnetic properties, the easternmost of the three Laanila dykes represents a deeper crustal level than the other two dykes studied. Palaeomagnetic and aeromagnetic data suggest that the Palaeoproterozoic granulite block containing the easternmost Laanila dyke was uplifted after the dykes were emplaced.

The Sm–Nd isotopic age determinations yield an age of 1042 ± 50 Ma for the Laanila dykes, 1013 ± 32 Ma for the Ristijärvi dykes and 1066 ± 34 Ma for the Kautokeino dykes indicating that the dyke swarms are of similar age within error limits. However, palaeomagnetic pole of the Laanila–Ristijärvi dykes (P_{lat} = 2.1 °S, P_{long} = 212.2 °E, dp = 12.7°, dm = 21.1°, N = 3 dykes) differs significantly from the pole of the Kautokeino dykes (Plat = 35.2 °S, P_{long} = 236.8 °E, N = 2 dykes). The isotope and palaeomagnetic ages of the Laanila–Ristijärvi dykes are considered agreeable, but on the basis of younger precisely dated Sveconorwegian palaeopoles, a magnetization age of *c.* 900 Ma for the Kautokeino dykes is inferred. Three explanations for the discrepancy are suggested: (i) local tectonic movements, (ii) delayed magnetization of the Kautokeino dykes with respect to that of the Laanila–Ristijärvi dykes, (iii) the isotope ages and palaeomagnetic ages are compatible but the Fennoscandian APWP should be revised.

The northern Fennoscandian shield (Fig. 1) is mainly defined by Archaean to Palaeoproterozoic rocks about 3.1–1.7 Ga old. However, igneous activity has taken place since cratonization, as manifested by the roughly 1.0 Ga old dyke swarms, of which the Laanila dyke swarm (Pesonen *et al.* 1986) is a good example. The Laanila dykes were injected along a NE–SW trending fracture zone that partly follows the fracturing of the Palaeoproterozoic (1.95–1.8 Ga) Lapland granulite belt. In the Finnmark area, northern Norway, the Palaeoproterozoic Kautokeino and Karasjok greenstone belts are also cut by NE–SW trending dykes. On aeromagnetic maps both the Laanila and Kautokeino dykes are shown as strong negative anomalies.

One of the topics of the present study is to define the primary remanent magnetization of the Laanila–Ristijärvi and Kautokeino dykes in order to better define the Fennoscandian Apparent Polar Wander Path (APWP) at Sveconorwegian time (*c.* 1100–850 Ma). Both dyke swarms are ideal for palaeomagnetic studies for three reasons. First, they carry stable remanence directions, which are not affected by later metamorphic events (Vihavainen 1981; Pesonen *et al.* 1986). Second, the dykes are exposed along their contacts, thus allowing baked contact tests to be made in order to establish primary intrusion magnetization. Third, they have been dated isotopically with the Sm–Nd method. Another topic is the correlation of the Laanila and Kautokeino dyke swarms in order to investigate the extent to which Neoproterozoic dyke injection took place in northern Fennoscandia. This is done by using the rock magnetic, petrophysical and petrographic properties of the dykes.

Geological setting

Laanila–Ristijärvi dykes

The Laanila diabase dyke swarm extends for over 100 km from Laanila to Ristijärvi in Finland and into northeastern Norway (Fig. 1). The dykes trend NE–SW in an en echelon-type manner. In the southwest, in Laanila, the dykes cut the Palaeoproterozoic Lapland granulite belt and in the northeast, in Ristijärvi, they cut the Archaean

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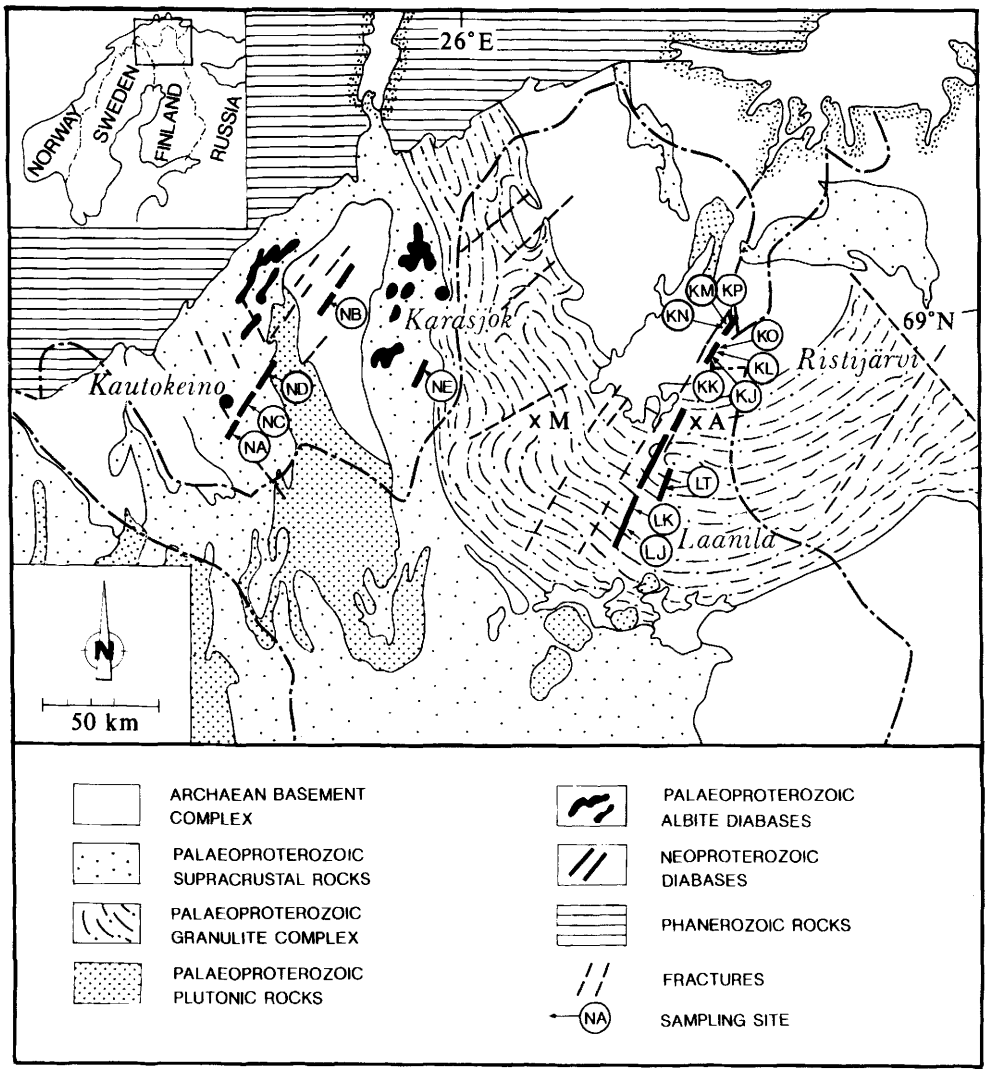


Fig. 1. Generalized geological map of the northern Fennoscandian Shield. A and M denote sampling sites at Akujärvi and Menesjärvi.

basement. The granulite belt is a high-grade metamorphic belt overthrust onto the Karelian basement in the southwest. U–Pb zircon analyses on the meta-igneous rocks of the granulite belt yield ages of 1.95–1.8 Ga, and the Archaean basement surrounding the Lapland granulite belt shows ages of 3.1–2.5 Ga (Meriläinen 1976).

The dyke swarm follows one of the main NE–SW directions of the extensive fracture zone in northern Fennoscandia. The fracture zone has been affected by long lasting repeated magmatic activity, as revealed by the roughly 1.8 Ga old post-orogenic granitoids (e.g. Haapala *et al.* 1987) and 300 Ma old diabase dykes on the Kola Peninsula (Gorbatschev *et al.* 1987; Fedotov & Amelin 1992), some 50 km from the area of our study. According to Pihlaja (1987) and Mutanen (pers. comm., 1992), the Laanila dykes represent steep fractures of the crust, extending for hundreds of metres to kilometres into the crust. The range of dyke width is in the Laanila area 1–20 m and in the Ristijärvi area 150–200 m. On aeromagnetic maps, the Laanila–Ristijärvi dykes are seen as strong negative anomalies. In chemical composition, the Laanila–Ristijärvi dykes consist of within-plate, rift-related tholeiitic basalt (Pihlaja 1987).

The diabase in Ristijärvi is an unmetamorphosed, homogeneous, medium-grained rock with ophitic texture. The grain size is fine at the margins and aphanitic right at the contact. The main minerals are plagioclase (andesine–labradorite), clinopyroxene and serpentine, with the latter having completely replaced olivine (see also Pihlaja 1987). The accessories are apatite, titanite, pyrite, pyrrhotite and magnetite. Sericite, chlorite and amphibole occur as alteration minerals and pyrite and magnetite as skeletal grains. In Laanila dyke LJ (Fig. 1), Mutanen (pers. comm., 1979) reports ilmenomagnetite composed of magnetite with ilmenite lamellae. The magnetite is partly altered to leucoxene and in dyke LJ the ilmenomagnetite is hypidiomorphic, not skeletal as in the Ristijärvi dyke. In Laanila dyke LT, the titanomagnetite is hypidiomorphic and skeletal, and has partly altered to leucoxene. Ilmenite occurs as platy separate grains. Dyke LT is better preserved than dyke LJ. However, the alteration is rather irregular from one site to another and is always related to the deuteric alteration associated with the latest phase of magma crystallization.

Kautokeino dykes

Palaeomagnetic investigations were conducted on NE–SW trending basaltic diabase dykes in the Kautokeino area of Finnmark, northern Norway (Fig. 1). Negative aeromagnetic anomalies (Olesen & Sandstad 1993) distinguish these dykes from the older (1815 ± 24 Ma, U–Pb zircon date, Krill *et al.* 1985) albite diabases, which, although similar in orientation (Fig. 1), have positive aeromagnetic anomalies. The Kautokeino dykes cut the Palaeoproterozoic Kautokeino greenstone belt and adjacent Archaean gneisses sharply (Olesen & Solli 1985; Olesen & Sandstad 1993). The Kautokeino greenstone belt forms a 40–80 km wide NNW–SSE trending rift-related synclinorium separated by steeply dipping faults from the Archaean Ráiseatnu Gneiss Complex in the west and the Jergol Gneiss Complex in the east. The area is cut by the NE–SW trending, 230 km long Mierujávri–Sværholt Fault Zone (MSFZ). Since the emplacement of the Kautokeino dykes, the greenstone belt has been tectonically active. The eastern fault of the belt shows sinistral displacement cutting the Kautokeino dykes and causing an offset of 1 km along the fault (Olesen & Sandstad 1993).

The Kautokeino dykes can be followed for about 50 km on low altitude aeromagnetic maps. The length of different dykes ranges from hundreds of metres to tens of kilometres; their width is 20–50 m and their dip varies from 60 to 70° SE. The mineral composition of the Kautokeino diabases corresponds to that of the Laanila diabases. The main minerals are plagioclase, which is is highly altered to sericite and saussurite, and clinopyroxene. Olivine phenocrysts are totally replaced by serpentine and chlorite. The main accessories are biotite, amphibole, magnetite and pyrrhotite. The magnetite occurs partly as skeletal and partly as subhedral or very small grains. The texture is ophitic.

Karasjok dykes

Midtun (1988) reports at least two SW–NE striking diabase dykes cutting the Karasjok greenstone belt some 80 km from the Kautokeino greenstone belt. These dykes, too, show up as strong negative aeromagnetic anomalies. Two pilot samples (provided by O. Olesen) from one of the dykes (NE, Fig. 1) were measured at the Geological Survey of Finland (GSF) and will be reported in this study.

Sampling

Laanila and Ristijärvi dykes

Palaeomagnetic investigations on the Laanila dykes started in 1978 (Vihavainen 1981; Neuvonen, pers. comm., 1986; Pesonen *et al.* 1986), when the first samples were taken from two Laanila dykes and host rocks at sites LJ, LK and LT (see Tables 2 and 4). In 1987, more samples were taken at site LK. Sites LJ and LK (hereafter called dyke LJ) are exposures of the same dyke, from which altogether 30 samples were taken. Five samples were taken from the host granulite at dyke LJ. Dyke LT, where eleven samples were taken, is located about 20 km east of dyke LJ. At site LT, four samples were taken from the host granulite. In 1988, the investigations were extended to Ristijärvi (close to the Russian border), which is a continuation of the Laanila dyke. At Ristijärvi (dyke RL), 26 diabase samples were taken from five sites; 13 samples were taken from host amphibolite at two sites at the contact and one sample was taken from the host Archaean gneiss.

Kautokeino and Karasjok dykes

In the Kautokeino area, diabase samples were taken at two sites (NA and NB, Fig. 1). At site NA, ten samples were taken from the diabase dyke and four samples at varying distances from the host Palaeoproterozoic amphibolite (see Tables 3 and 5). Site NB lies about 30 km northeast from site NA and cuts the Archaean Jergol gneiss. Nine samples were taken from the diabase and four from the gneiss. In addition, O. Olesen (of Geological Survey of Norway) provided three pilot samples from site NB, two from sites NC and ND (Fig. 1) in

the Kautokeino area and two from a diabase dyke (NE; Fig. 1) in the Karasjok area.

Sm–Nd Ages

Isotopic dating of the dykes involved Sm–Nd analyses on whole rocks and concentrates of fresh magmatic clinopyroxene and slightly altered plagioclase. The analyses have been performed in three stages: (1) old analyses on the Laanila dyke using non-commercial mass-spectrometer (Huhma 1986); (2) analyses using VG SECTOR 54 with old Faraday buckets (before Aug. 94); and (3) recent analyses after Faraday bucket replacement. We have noticed that analyses with old buckets can be seriously biased, even after standard correction (cf. Thirlwall 1991). Therefore we have not used those data for age determination.

The whole rocks and clinopyroxenes from the Laanila dyke give a Sm–Nd age of 1042 ± 50 Ma with an initial epsilon of +4.8 ± 0.5 (n = 5, MSWD = 1.4). The new data on the Ristijärvi dyke provide an age of 1013 ± 32 Ma ε_{Nd} = +4.8, n = 3). Due to poor analyses on plagioclase (too small sample size) this age is in fact based only on whole rock and pyroxene analyses. All nine analyses listed in Table 1 (and thus including also potentially biased data) give an age of 1005 ± 43 Ma (n = 9, MSWD = 4). The five new analyses on the Kautokeino dyke give an age of 1066 ± 34 Ma

Table 1. *Sm–Nd data on whole rocks and mineral concentrates.*

Sample	Fraction	Sm (ppm)	Nd (ppm)	$^{147}Sm/^{144}Nd$	$^{143}Nd/^{144}Nd \pm 2\ \sigma m$
Laanila diabase, samples LJ10 (= A540) and LT2:					
[2]LJ10 wr		5.58	17.71	0.1905	0.512835 ± 34
[2]LJ10 cpx	hp	2.43	5.13	0.2870	0.513457 ± 56
[2]LT2 wr		4.49	14.25	0.1905	0.512857 ± 28
[2]LT2 cpx	hp	3.97	9.01	0.2667	0.513363 ± 32
LJ10 cpx#2	hp	2.39	4.90	0.2944	0.513563 ± 10
Ristijärvi diabase, sample A1227 (= KJ2-1B/4821):					
[1]wr		5.47	17.05	0.1941	0.512885 ± 20
[1]wr#2		5.29	16.44	0.1945	0.512898 ± 20
[1]pl	2.62–2.76/nhp	0.18	0.75	0.1414	0.512580 ± 30
[1]pl#2	2.62–2.76/hp	0.17	0.74	0.1423	0.512603 ± 90
[1]cpx	+3.4/m 0.6A/nhp	3.74	7.60	0.2974	0.513600 ± 20
[1]cpx#2	3.4/m 0.6A/hp	3.36	6.79	0.2994	0.513575 ± 20
wr#3		5.31	16.61	0.1931	0.512865 ± 10
pl#3	2.62-2.76	0.21	0.87	0.1486	0.512576 ± 133
cpx#3	+3.4/m 0.6A/nhp	3.90	8.21	0.2877	0.513494 ± 10
Kautokeino diabase A1228 (= NA2-1E/2823):					
[1]wr		5.26	15.30	0.2078	0.513030 ± 20
[1]pl	2.65-2.67/+100/hp	0.94	3.41	0.1663	0.512789 ± 20
[1]cpx	3.3–3.4/m 0.6A/hp	2.83	5.77	0.2970	0.513603 ± 20
wr#2		5.18	15.13	0.2070	0.513023 ± 20
pl#2	2.62–2.75/–160/nhp	0.71	2.46	0.1739	0.512775 ± 11
pl#3	2.62–2.75/–160/nhp	0.45	1.60	0.1693	0.512736 ± 10
cpx#2	3.3–3.4/m 0.6A/nhp	3.15	6.81	0.2795	0.513520 ± 16
cpx#3	3.3–3.4/m 0.6A/nhp	2.97	6.15	0.2923	0.513596 ± 10

Sample: wr = whole rock, pl = plagioclase, cpx = clinopyroxene, #2 = duplicate.
Fraction: density/m = magnetic/+160 = ∅>160 μm/ hp = handpicked, nhp = not handpicked.
Concentrations were determined using isotope dilution by addition of a mixed ^{149}Sm–^{150}Nd spike without aliquoting (except old analyses on the Laanila dyke/[2]). Based on several duplicated analyses in the laboratory the estimated error in $^{147}Sm/^{144}Nd$ is 0.4%. $^{143}Nd/^{144}Nd$ ratio is normalized to $^{146}Nd/^{144}Nd = 0.7219$. The new set of samples was analysed using VG SECTOR 54 with new Faraday buckets and the average value for La Jolla standard was $^{143}Nd/^{144}Nd = 0.51185 \pm 1$ (SD, n = 20, errors in last significant digits).
[1] = Accuracy problems due to old Faraday buckets in VG SECTOR 54 (analysed before Aug. 94). Based on frequent analyses on La Jolla standard these data should be compatible with La Jolla value $^{143}Nd/^{144}Nd = 0.51185$. The data with new collectors show that this correction was not applicable for all analyses. These data are not shown in figures nor used for age determinations.
[2] = Old analyses using non-commercial mass-spectrometer (for methods see Huhma 1986).

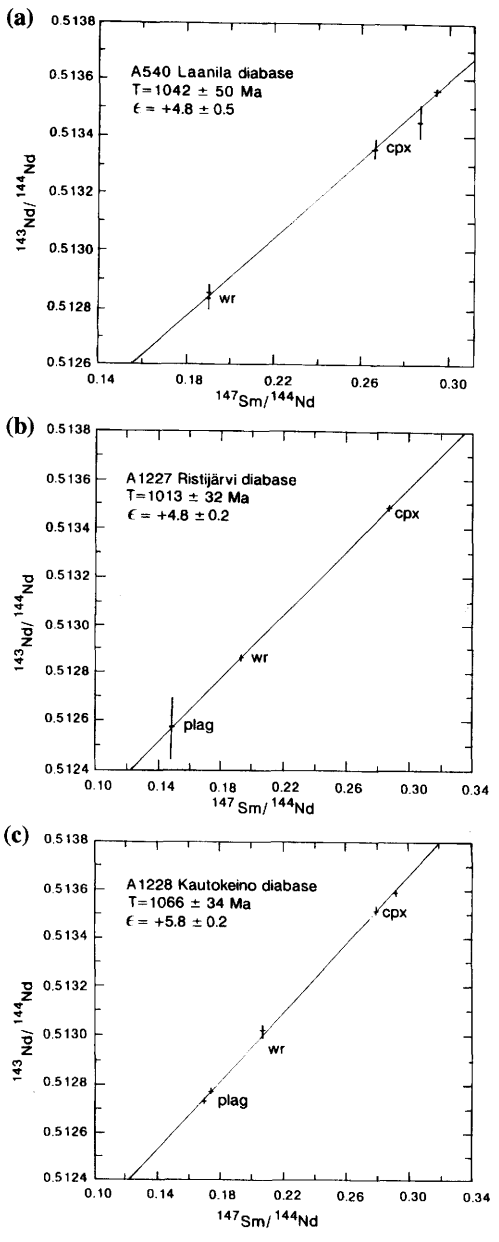


Fig. 2. Sm-Nd diagrams for (**a**) Laanila dykes, (**b**) Ristijärvi dykes and (**c**) Kautokeino dykes.

ε_{Nd} = +5.8 ± 0.2, MSWD = 2). The old analyses on plagioclase and pyroxene are off the line.

The Sm–Nd ages of the dykes analysed are thus indistinguishable within error. The nearly chondritic Sm/Nd ratios and positive ε_{Nd} values suggest that the mafic magma originated from depleted mantle reservoirs. The ε_{Nd} for the Kautokeino dyke is close to the depleted mantle of DePaolo (1981), whereas ε_{Nd} for the Ristijärvi and Laanila dykes is slightly lower suggesting another source or some crustal contamination.

Palaeomagnetic measurements and analysis

The remanences were measured using a spinner magnetometer built at the Geological Survey of Finland (GSF) (see Pesonen *et al.* 1983). About half of the specimens from each core were demagnetized with an alternating field (AF) using a Schonstedt or a homebuilt AF demagnetizer. The other half were thermally demagnetized by a Schonstedt thermal demagnetizer. In AF demagnetization, most specimens were demagnetized in 12 steps of 2.5, 5, 10, 20, 30, 40, 50, 60, 70, 80, 90 and 100 mT. Some samples were AF demagnetized up to 110 mT, some only up to 40–60 mT. In thermal demagnetization, temperatures of 200, 300, 350, 400, 500, 520, 540, 560, 570, 580, 600, 620, 640, 660 and 680°C were used. To establish whether mineralogical changes occurred upon heating, bulk susceptibility was measured after demagnetization at temperatures of 200, 300, 400, 500, 570, 600, 620, 640 and 680°C. A joint analysis of stereographic plots, demagnetization decay curves, orthogonal demagnetization diagrams (Zijderveld 1967; Leino 1991) and principal component analysis (Kirschvink 1980) were used to obtain the remanence directions. These were accepted if the angular deviation (MAD) was $\leq$ °6. Mean remanence directions were calculated using Fisher (1953) statistics.

To distinguish the remanence carriers, magnetic hysteresis measurements were conducted on nine specimens (at the Technical Research Centre of Finland) representing characteristic remanence behaviour of the different sites. Another seven specimens were subjected to saturation isothermal remanent magnetization (SIRM) and anhysteretic remanent magnetization (ARM) analysis for conducting Lowrie–Fuller tests. Magnetic minerals were identified measuring thermomagnetic curves (low-field susceptibility versus temperature) on five specimens using a KLY-2 kappabridge connected to a CS-2 furnace (Geofyzika Brno). The measurements were made at a temperature range of 30–700 °C.

Palaeomagnetic results

Laanila–Ristijärvi dykes

According to their palaeomagnetic directions, the sampled Laanila–Ristijärvi dykes represent three

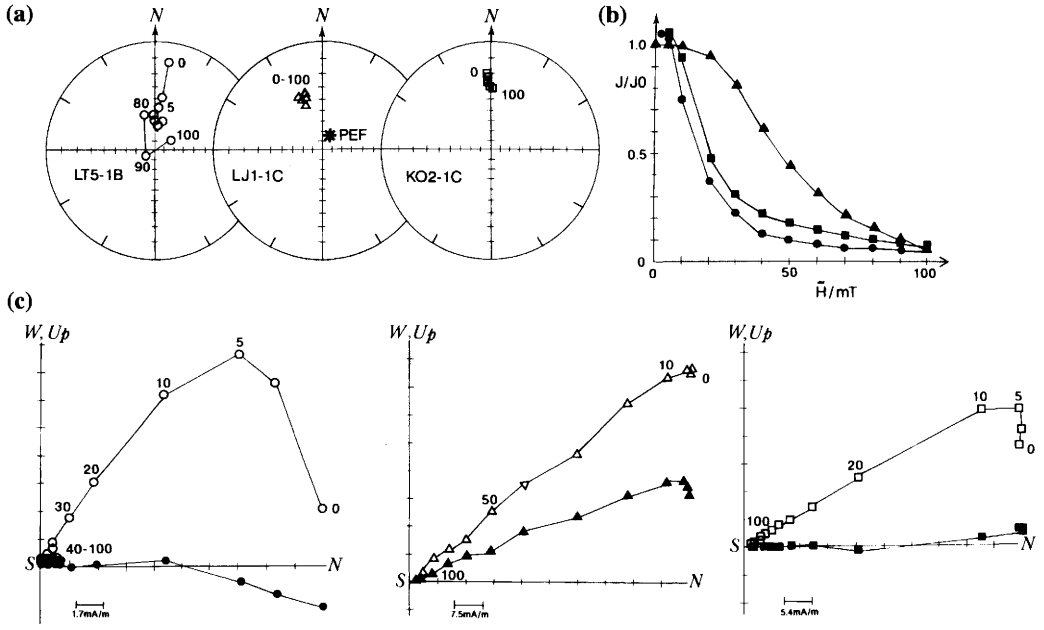


Fig. 3. Examples of AF demagnetization behaviour typical of Laanila–Ristijärvi dykes. (**a**) Stereographic projections of directional data on demagnetization, where PEF shows direction of present Earth's magnetic field; (**b**) relative intensity decay curves; (**c**) orthogonal demagnetization diagrams. Open (closed) symbols denote vertical (horizontal) planes. Numbers by each demagnetization step denote peak alternating field (mT).

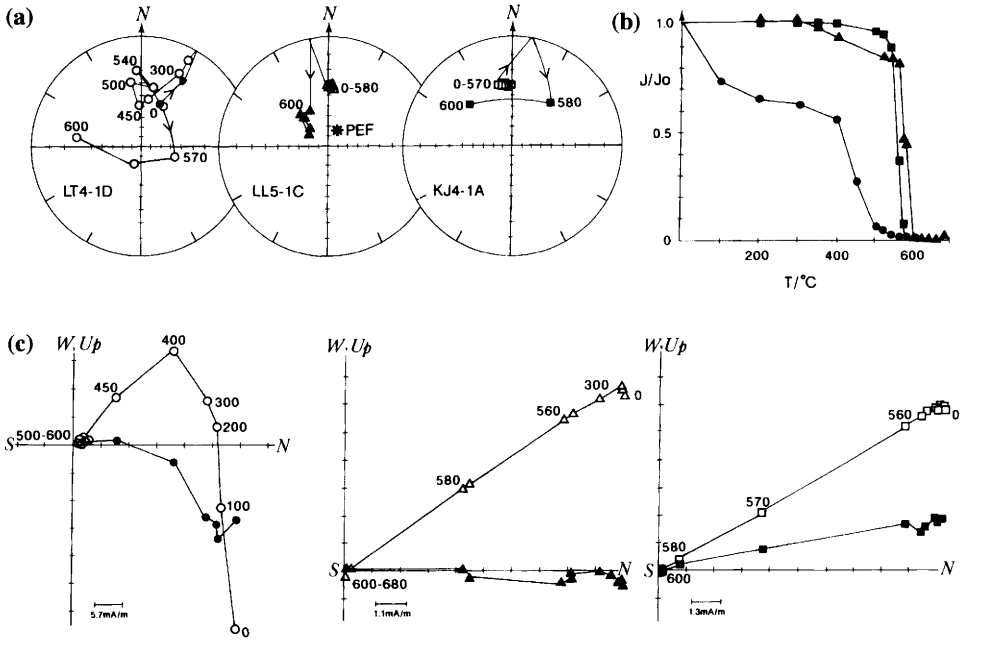


Fig. 4. Examples of thermal demagnetization behaviour typical of Laanila–Ristijärvi dykes. (**a**) Stereographic projections; (**b**) relative intensity decay curve; (**c**) orthogonal demagnetization diagrams. Numbers by each demagnetization step denote temperature (°C). Other symbols as in Fig. 3.

Table 2. *Mean palaeomagnetic data on Laanila–Ristijärvi dykes and host rocks*

Site/Dyke	*B/N/n*	D (°)	I (°)	α_{95} (°)	k	P_{lat} (N°)	P_{long} (E°)	dp (°)	dm (°)	(A_{95}) (°)
				Laanila						
Diabase										
Mean LT (68.5, 28.15)	1/*9/33	357.3	–52.5	8.7	36.3	–11.6	210.4	8.2	11.9	
LK	*15/57	350.7	–34.4	3.8	100.9	2.4	216.3	2.5	4.4	
LJ	*13/37	350.3	–34.4	7.7	30.1	2.4	216.7	5.0	8.8	
Mean LJ (68.4, 27.5)	*2/28/84	350.5	–34.4	—	—	2.4	216.5	—	—	
				Ristijärvi						
KJ	*5/17	355.6	–30.3	10.3	56.4	4.7	213.1	6.4	11.4	
KK	*1/3	355.5	–31.1	—	—	4.1	213.2	—	—	
KO	*5/15	1.7	–30.8	2.6	675.6	4.4	207.3	1.6	2.9	
KL	*6/11	4.1	–40.0	5.6	145.0	–1.8	205.2	4.0	6.7	
KN[1]	*4/10	254.8	25.4	14.7	40.3	7.1	317.8	8.5	15.8	
Mean RL (69.0, 28.9)	*4/17/46	359.1	–33.1	6.7	189.9	2.9	209.8	4.3	7.6	
Grand mean (68.7, 28.1)	*3/7/54/163	355.5	–40.0	17.5	50.5	–2.1	212.2	12.7	21.1	(13.8)
				Host rocks						
				Laanila						
Granulite										
LJ8-9, contact	*2/5	7.3	–33.3	—	—	3.2	200.6	—	—	
LJ19, 7 m	*1/4	21.3	–37.9	—	—	–1.0	187.7	—	—	
LT10, contact	*1/1	88.5	–50.6	—	—	—	—	—	—	
				Ristijärvi						
Amphibolite										
KM2–6, contact	*5/10	344.0	–16.6	8.4	84.5	11.7	225.1	4.5	8.6	
Gneiss										
KL11, contact	*1/3	27.3	–10.2	—	—	13.5	180.9	—	—	

NOTE: Numbers below the name of the dyke denote geographical latitude (°N) and longitude (°E) of the sampling site; for the host rocks, distance from the dyke contact is shown; *B/N/n* is the number of sites/samples/specimens used in statistical calculations; * denotes the statistical level used in mean calculations; [1] not included in mean direction; D and I are mean declination and inclination, respectively; α_{95} is the radius of the circle of 95% confidence; k is Fisher's (1953) concentration parameter; P_{lat} and P_{long} are the palaeolatitude and palaeolongitude for the virtual geomagnetic poles (VGPs) of different sites; the mean pole is the mean of VGPs; dp and dm are the semi–axes of the oval of 95% confidence for each VGP; (A_{95}) is the radius of the circle of 95% confidence of the mean VGP poles. All poles are of normal (N) polarity.

separate dykes: LT and LJ (in the Laanila area) and RL (in the Ristijärvi area). Examples of alternating field (AF) and thermal demagnetizations of the dykes are given in Figs 3 and 4. The dykes carry a coherent northward characteristic remanent magnetization (ChRM) with moderate negative inclination that is most stable and best isolated at sites LJ and LK in Laanila and at site KJ in Ristijärvi (Table 2). In addition, a steep positive component pointing northward was isolated in all specimens at low coercivities and low unblocking temperatures. The direction of this component is close to the present Earth's magnetic field (PEF) and is therefore interpreted as representing recent viscous remanent magnetization (VRM). Most specimens exhibit consistent behaviour upon AF and thermal demagnetization, indicating that both cleaning methods effectively remove the PEF and isolate the ChRM. In some samples, however, AF cleaning was more effective at isolating the ChRM from the PEF component.

Dyke LT has a very large contribution of the present Earth's magnetic field (PEF) component. The inclination of the natural remanent magnetization (NRM) direction before demagnetization is therefore positive. In dykes LJ and RL, the relative amount of PEF is lower and the inclination of NRM is negative at most sites. The mean inclination of the characteristic high coercivity remanence component at site LT is about –50° (Table 2), which is at a high angle (almost opposite) to the PEF direction. In the two other dykes, the inclination is shallower (Table 2) and could be due to incomplete cleaning, the component contaminated by PEF magnetization, which would make the upward characteristic component shallower. However, because both components are sharply isolated in orthogonal plots (Fig. 3) the inclination difference is probably not due to incomplete demagnetization but is real.

Thermal demagnetization decay curves for specimens of dyke LT indicate that the remanence carriers have more widely distributed unblocking temperatures than do those of the two other Laanila dykes (Fig. 4), clearly owing to the larger grain size distribution of the LT specimens. This is consistent with petrophysical and hysteresis properties as will be shown later. The scatter of remanence directions in dyke LT is comparatively high. The two other Laanila dykes show narrow unblocking temperature spectra, the bulk being unblocked between 560° and 600° C indicative of Ti-poor magnetite. The clear change in the slope of the thermal demagnetization curve at about 350° C in some samples may be due to pyrrhotite.

The mean remanence directions at sites LJ and LK in Laanila dyke LJ are almost identical (Table 2) confirming that the sites represent exposures of the same dyke. The remanence direction of Ristijärvi dyke RL agrees with that for dyke LJ of Laanila within the α_{95} confidence level (Fig. 5). In contrast, the inclination of dyke LT is about 20° steeper than that of dykes LJ or RL. The between-dyke scatter may be due to secular variation, but the possibility of tectonic influence or a slight age difference between dykes LT and LJ–RL cannot be ruled out, as will be discussed later. The mean direction and the corresponding palaeomagnetic pole, calculated for a common reference locality, are averaged for the LT, LJ and RL dykes (Table 2, Fig. 5).

Kautokeino dykes

The characteristic remanent magnetization (ChRM) of the Kautokeino dykes after demagnetization has a steep northward negative inclination, typically reached as a stable end point at 100 mT (Fig. 6). All samples also exhibit a pronounced steep downward direction that is removed in low AF fields (5–30 mT). Since it is aligned approximately along the present Earth's field direction (being almost antiparallel to the characteristic remanence direction), it is most likely a viscous remanent magnetization of recent origin. Depending on the magnitude of the PEF component, the intensity decay curves and orthogonal plots behave very

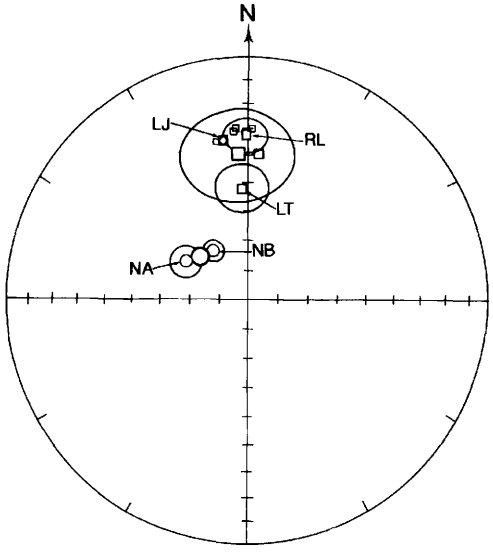


Fig. 5. Mean remanence directions and α_{95} confidence circles for Laanila–Ristijärvi (squares) and Kautokeino (circles) diabase dykes. Smallest symbols denote site means, intermediate symbols dyke means and largest symbols dyke swarm mean directions.

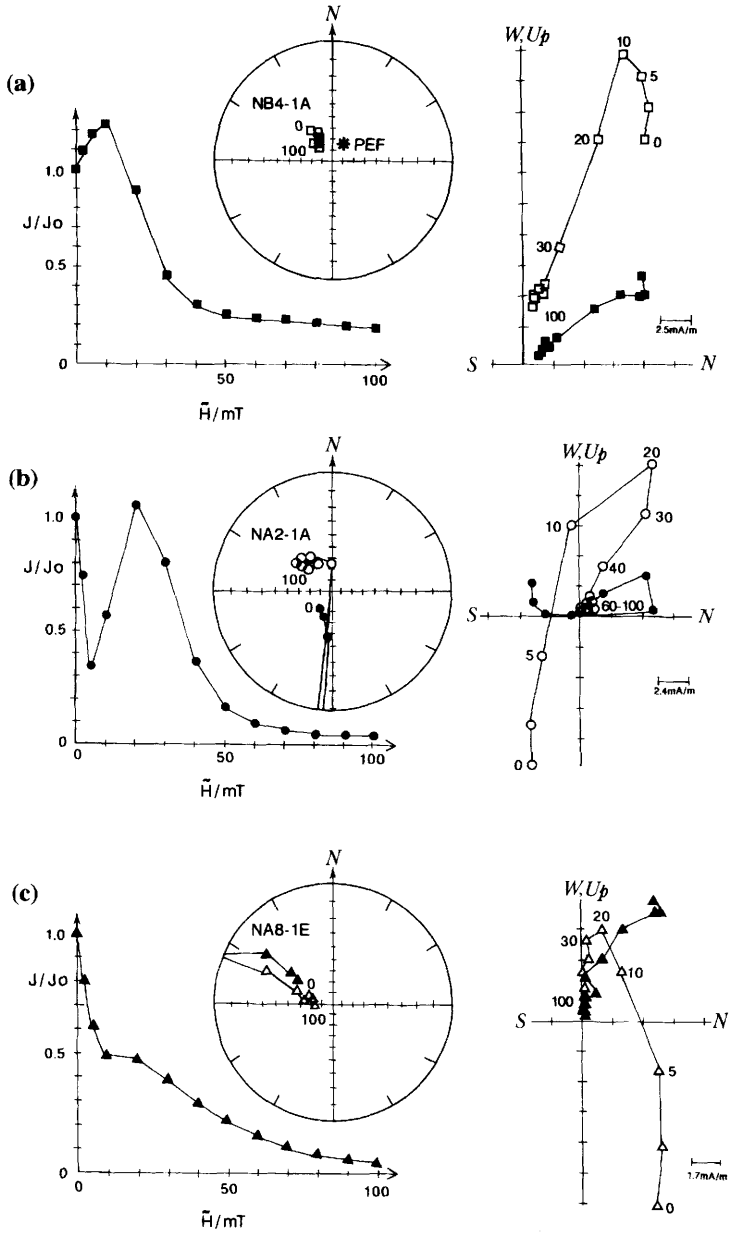


Fig. 6. Examples of AF demagnetization behaviour for the Kautokeino dykes NA and NB. Symbols as in Fig. 3.

differently (Fig. 6a–c) in AF demagnetization. In some of the samples, the magnitude of PEF is so high that the NRM direction (before demagnetization) has a steep to moderate positive inclination. Thermal demagnetization (Fig. 7) does not always isolate the ChRM as effectively as AF demagnetization, and the data are more scattered. Laboratory unblocking temperatures of 520–580° C indicate that titanomagnetite is the remanence carrier.

The mean remanence directions and their statistics are summarized in Table 3. The dyke mean is averaged both for the two dykes (NA and NB) and for all samples. The difference in directions between dykes is small and can be attributed to secular variation.

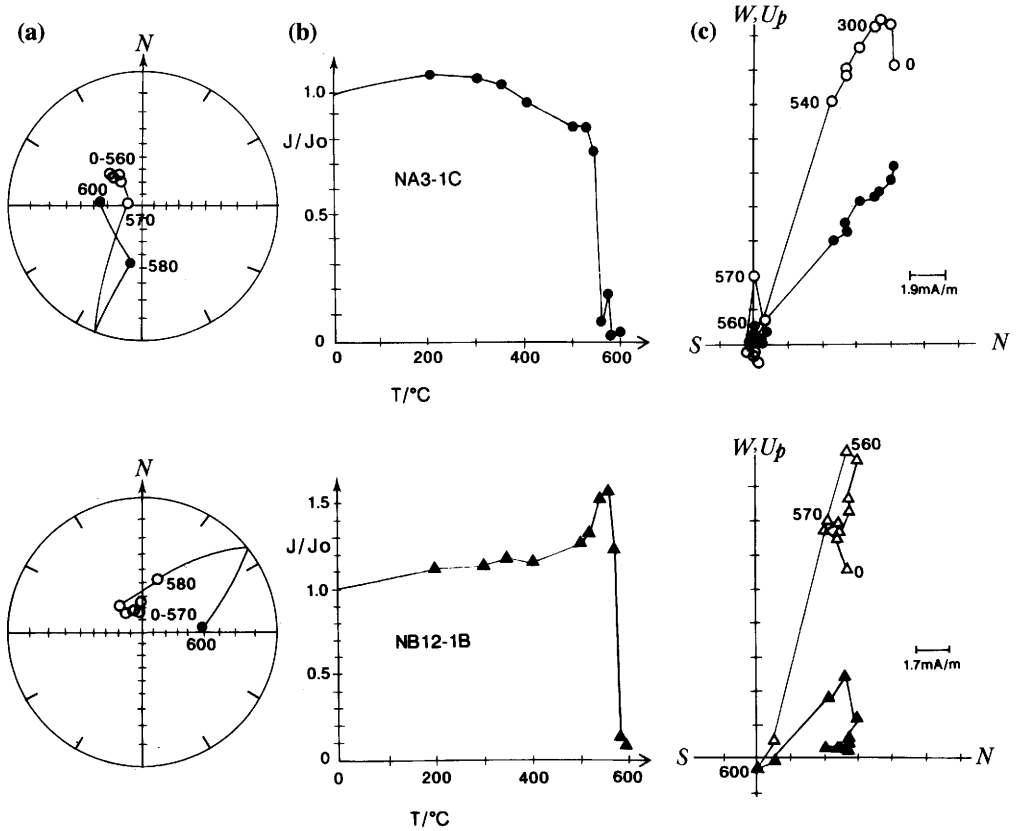


Fig. 7. Examples of thermal demagnetization behaviour typical of Kautokeino dykes. Symbols as in Figs 3 and 4.

Palaeomagnetic data on two pilot samples taken from two other sites of the Kautokeino dyke are also shown in Table 3. They are not included in the statistics because only one sample was measured from both dykes. The remanence directions agree well with those at sites NA and NB. Sample ND was taken from the eastern side of the sinistral fault transecting the diabase dyke (Olesen & Sandstadt 1993), and site NA from the western side of the fault. The remanence directions are identical within the limits of error and, thus, no fault movement can be observed in the palaeomagnetic data. Olesen & Sandstadt (1993) also report NRM results for the Kautokeino dyke, obtaining a direction ($D = 336°$, $I = -56°$, $\alpha_{95} = 9°$, $n = 11$ samples) consistent with our results, but shallower inclination (12°), presumably due to the large, not erased PEF component (see also pole OS in Fig. 19).

Karasjok dykes

Palaeomagnetic data on two pilot samples from one diabase dyke in the Karasjok area are also given in Table 3 and they are in good agreement with the Kautokeino data. The NRM direction ($D = 315°$, $I = -63°$, $\alpha_{95} = 7.1°$, $n = 6$ samples) reported by Midtun (1988) for two more diabase dykes in the Karasjok area is also comparable to the mean remanence direction (see also pole MI in Fig. 19) of the Kautokeino dykes. The agreement between the data on the Karasjok dykes, which are located between Kautokeino and Laanila, indicates that dykes with a similar reversed magnetization direction occur over a wide area in northern Fennoscandia and that they are probably contemporary in age and can thus be correlated.

Bulk petrophysics

Petrophysical properties (susceptibility, remanence intensity and density) were studied to correlate different dykes within a dyke swarm (e.g. Laanila) and between dyke swarms (Laanila vs. Kautokeino). Petrophysical properties are informative about the remanence stability, mineralogy and grain size of the rocks. A susceptibility versus

Table 3. *Mean palaeomagnetic data on Kautokeino and Karasjok dykes and host rocks*

Dyke	*B/N/n*	D (°)	I (°)	α_{95} (°)	k	P_{lat} (N°)	P_{long} (E°)	dp (°)	dm (°)	(A_{95}) (°)
					Kautokeino					
Diabase										
NA (69.0, 23.3)	1/*9/29	306.2	–67.4	5.2	97.9	–35.5	242.8	7.2	8.7	
NB (69.2, 23.8)	1/*12/19	322.3	–68.8	3.6	150.8	–34.5	230.9	5.1	6.0	
Mean	*/2/21/48	314.0	–68.3	—	—	–35.2	236.8	—	—	—
Mean (69.1, 23.6)	2/*21/48	315.1	–68.4	3.1	108.6	–35.6	235.7	3.1	5.4	(4.7)
NC[1] (69,0, 23.4)	*1/1	328.0	–56.2	—	—	–18.4	230.0	—	—	
ND[1] (69.0, 23.4E)	*1/1	309.5	–70.3	—	—	–38.83	238.6	—	—	
					Karasjok					
NE[2] (69.1, 25.2)	2/*2	324.0	–65.1	—	—	–29.2	232.4	—	—	
					Host rocks					
					Kautokeino					
Amphibolite										
NA11, contact	*1/2	212.5	–76.1	—	—	–76.1	299.8	—	—	
NA12, 17 m	*1/2	2.2	–53.9	—	—	–13.4	201.4	—	—	
Gneiss										
NB2, contact	*1/1	351.8	–57.6	—	—	–17.6	210.6	—	—	
NB14, 1 m	*1/1	57.7	–31.2	—	—	—	—	—	—	
NB15, 20 m	*1/1	226.5	62.7	—	—	—	—	—	—	

[1] pilot samples from two additional diabase dykes at Kautokeino , [2] pilot samples from one diabase dyke at Karasjok; [1] and [2] are not included in mean direction. For other explanations, see Table 2.

NRM intensity plot is shown in Fig. 8 and a susceptibility versus density plot in Fig. 9. The results are summarized in Tables 4 and 5. The petrophysical properties of the host rocks are discussed in the context of baked contact test.

Laanila–Ristijärvi dykes

There is only little variation in bulk petrophysical properties between different Laanila–Ristijärvi dykes LT, LJ and RL (Fig. 8). On average, the samples carry comparatively high Q values between about 2.5 and 10 (Table 4), indicating that the dykes have stable remanences probably carried by single-domain (SD)/pseudo-single-domain (PSD) magnetite grains.

In dyke LT, the susceptibility is somewhat higher and the NRM intensity and Q values are lower than in dykes LJ or RL. This is obviously due to the large contribution of the VRM component, which reduces the total resultant NRM intensity. The lower NRM and Q values in LT support the observation from demagnetization data that the remanence in LT is carried by larger multidomain (MD) grains, which are therefore less stable and more likely to acquire viscous remanence (e.g. Dunlop 1983) than the grains in other dykes. Unlike dykes LJ and RL, dyke LT does not show up on the aeromagnetic maps because the positive remanence inclination (due to PEF contamination) makes the magnetic anomaly of this site comparable to the anomalies typical of the host granulite. Dykes LJ and RL are seen as strong negative anomalies due to the dominance of the negative inclinations of NRM, with Q values up to 10.4 (Table 4). The very high Q values (> 18) found in three KK samples are probably caused by lightning-induced isothermal remanent magnetization (IRM).

The variation in density between dykes LT, LJ and RL is more pronounced than the variation in NRM (Fig. 9a, Table 4). Dyke LT, characterized by the highest susceptibilities, has the lowest density, which may be due to a higher proportion of titanomagnetite at the expense of pyroxenes. The highest densities occur at Ristijärvi site KL, obviously due

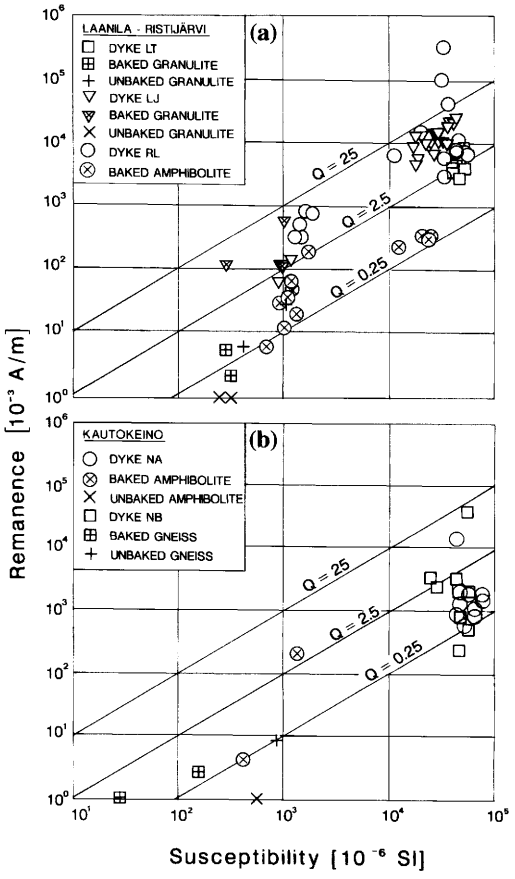


Fig. 8. Remanence–susceptibility diagrams for diabase dykes and host rocks from (**a**) Laanila–Ristijärvi and (**b**) Kautokeino. Koenigsberger's Q ratios are shown as lines.

to the high proportion of pyroxenes. Dykes LJ and RL show a clear duality in susceptibility that is not seen in dyke LT, possibly because the magma of dykes LJ and RL may have been more inhomogeneous than that of dyke LT.

Kautokeino dykes

The bulk susceptibility and remanence intensities are similar in dykes NA and NB (Fig. 8b, Table 5). Q values range between 0.25 and 2.5, and are an order of magnitude lower than in the Laanila–Ristijärvi dykes. This is partly explained by the strong VRM component in the PEF direction. In the Kautokeino dykes, the ChRM component is almost opposite the PEF which reduces the total intensity of remanent magnetization. However, in many samples the proportion of the VRM component is so small that the total remanence direction still retains the negative inclination. The uncleaned NRM direction obtained by Olesen & Sandstad (1993) is also close to the direction of ChRM, being steep and negative. Thus, the total NRM intensity, as well as the Q values, of the Kautokeino dykes are slightly lower than those of the Laanila dykes. In the density versus susceptibility plot (Fig. 9b), the densities show only little variation between dykes NA and NB. On average, the densities and susceptibilities accord with those of the Laanila–Ristijärvi dykes; the densities in particular are comparable to those of dyke RL but are slightly higher than those of dykes LT and LJ.

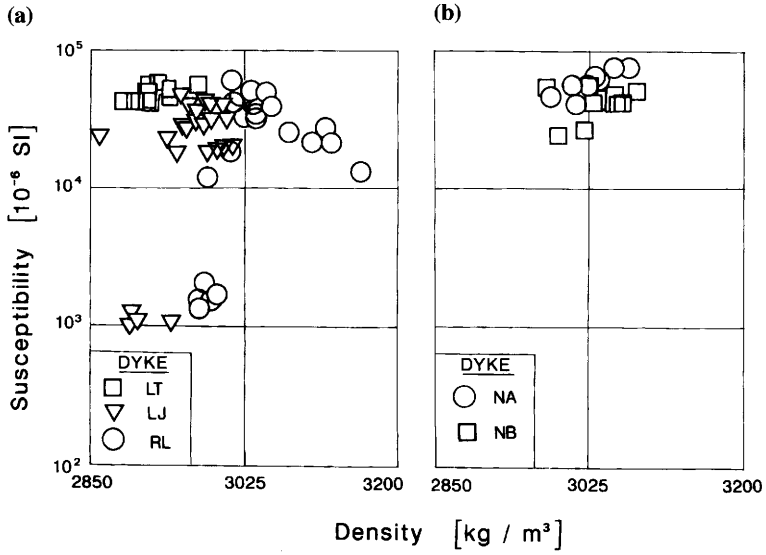


Fig. 9. Susceptibility–density diagrams for diabase dykes from (**a**) Laanila–Ristijärvi and (**b**) Kautokeino.

Table 4. *Petrophysical properties of Laanila–Ristijärvi dykes and host rocks*

Site/Dyke	*B/N/n*	*D*	*K*	*J*	*Q*
		Laanila			
Diabase					
Mean LT	1/*11/14	2920	47571	4705.3	2.5
LK	*16/34	2973	29725	11696.5	10.4
LJ	*14/15	2954	21876	11714.2	9.7
Mean LJ	*2/30/49	2963.5	25800.5	11705.4	10.1
		Ristijärvi			
KJ	*5/18	3017	27084	11076.2	8.4
KK	*4/13	3034	37650	[1]123096.8	[1]18.4
KO	*6/18	3016	41335	6304.9	3.9
KL	*6/11	2983	1621	610.4	9.4
KN	*5/12	3032	45947	7437.1	4.1
Mean RL	*5/26/72	3016	30727	[2]6357.2	[2]6.5
Grand Mean	*3/8/67/135	2966.5	34699.5	7589.3	6.4
		Host rocks			
		Laanila			
Granulite					
LJ7-9, contact	*3/3	2843	787	257.7	8.4
LJ19-20, 7 m	*2/2	2719	288	1.0	0.1
LT10-11, contact	*2/2	2653	315	3.5	0.3
LT13-14	*2/3	2682	798	16.0	0.5
		Ristijärvi			
Amphibolite					
KM1-6, contact	*6/17	3082	1214	59.1	0.8
KP1-7, contact	*7/13	3102	14577	2485.8	0.4
Gneiss					
KL11, contact	*1/4	2663	381	19.8	1.2

NOTE: For host rocks, the distance of the host rock from the dyke is shown; *B/N/n* is the number of sites/samples/specimens used in the statistical calculations; * denotes the statistical level used in mean calculations, *D* is density (kg m^{-3}); *K* is susceptibility (10^{-6} SI); *J* is remanence intensity (mA m^{-1}); $Q = J/(K \times H)$ where *H* is the present Earth's magnetic field intensity. [1] not included in mean value, [2] *B/N/n* = *4/22/59.

Table 5. *Petrophysical properties of Kautokeino dykes and host rocks*

Site	*B/N/n*	*D*	*K*	*J*	*Q*
Diabase					
NA	1/*10/37	3028	55298	2673.4	0.7
NB	1/*10/33	3041	41977	1813.2	1.2
Mean	*2/20/70	3035	48638	2243.3	1.0
		Host rocks			
Amphibolite					
NA11, contact	1/*2	3101	1305	207.4	3.8
NA12, 17 m	1/*4	2919	406	3.8	0.2
NA13-14, 20 m	*2/7	2971	612	0.9	0.1
Gneiss					
NB2, contact	1/*2	2614	140	3.0	0.6
NB14, 1 m	1/*2	2594	28	0.7	1.6
NB15, 20 m	1/*2	2623	826	8.7	0.3

NOTE: For explanations, see Table 3.

Anisotropy of susceptibility

The anisotropy of susceptibility (AMS) was measured on about 30 samples (Table 6). One reason for AMS measurements was to investigate whether the scatter of remanence directions might be partly caused by AMS. The remanence directions may deviate from the ambient geomagnetic field direction owing to either later deformation or magma flow along dyke margins. Palaeomagnetic interpretations of ages and tectonics may therefore be erroneous. However, the AMS does not always indicate that the **remanence** has been deflected, because the remanence carriers may be different from the AMS carriers (e.g. Tarling & Hrouda 1993). There are both SD and MD grains in the Laanila and Kautokeino dykes, and so the larger grains (carrying the VRM component) are mainly responsible for the bulk susceptibilities. On the other hand, the small (SD/PSD) magnetite grains presumably carry the primary characteristic remanence and may not be affected by deflection due AMS (see e.g. Hyodo & Dunlop 1993). The AMS data will therefore mainly indicate the deflection of NRM due to large MD magnetite grains. Here we discuss AMS only. According to Puranen *et al.* (1992), AF demagnetization does not cause significant changes in AMS; hence measurements were conducted for both AF demagnetized samples and fresh specimens.

In the Laanila–Ristijärvi and Kautokeino dykes the degree of anisotropy (P, the ratio of maximum and minimum susceptibilities) is typically below 1.05 (Table 6), which is the limiting value above which the remanence may have been deflected (Stephenson *et al.* 1986). Hence, since the MD grains were not deflected due to AMS, the remanence directions in the Laanila–Ristijärvi and Kautokeino dykes record the ambient geomagnetic field and are suitable for palaeomagnetic and tectonic studies.

The L (lineation)–F (foliation) diagram of AMS data (Fig. 10) shows that both the magnetic foliation and lineation values are below 1.05, indicating that the fabric is due to magma flow and that the dykes have not been affected by later deformation (see e.g. Hrouda & Chlupácôvá 1980; Tarling & Hrouda 1993).

Another use of AMS data is for defining the direction of magma flow. When basic magma moves within the crust, the magnetite grains are aligned with the long axes parallel to the direction

Table 6. *Degree of anisotropy (P) in Laanila–Ristijärvi and Kautokeino dykes*

Laanila–Ristijärvi		*Kautokeino*	
Specimen	*P*	Specimen	*P*
LT1-1D	1.010	NA1-1A*	1.024
LT2-1D*	1.011	NA3-1A*	1.043
LT5-1C	1.014	NA3-1D	1.028
LT7-1C	1.015	NA8-1E*	1.009
LT12-1A*	1.005	NA9-1C	1.057
LJ4-1C	1.017	NA9-1E*	1.029
LJ11-1A	1.026	NB3-1A*	1.013
LJ15-2B	1.006	NB3-1B*	1.014
LL1-1E	1.038	NB5-1A*	1.013
LL4-1D	1.032	NB6-1A*	1.041
LL7-1A	1.019		
LK3-3C	1.019		
LK5-1F	1.031		
LK7-1F	1.039		
KJ4-1E	1.026		
KL14-1A*	1.007		
KL15-2A*	1.008		
KK4-1C*	1.044		
KO2-1B	1.021		
KO5-1A	1.018		
KN1-1A	1.021		
KN4-2B	1.038		

NOTE: P is the ratio between maximum and minimum susceptibility. * = specimen demagnetized with alternating field before anisotropy of susceptibility (AMS) measurement.

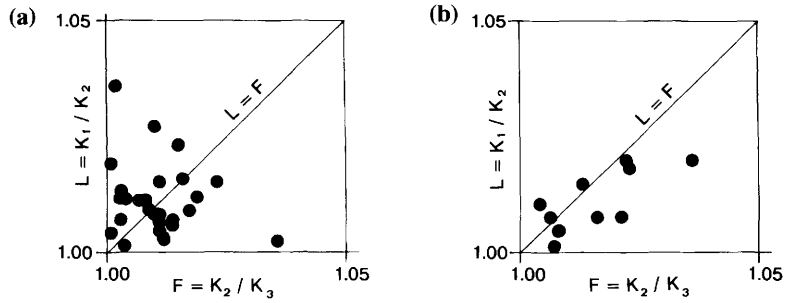


Fig. 10. Magnetic foliation (F) – lineation (L) diagram for (**a**) Laanila–Ristijärvi and (**b**) Kautokeino diabases. F and L values below 1.05 are characteristic of magmatic flow fabrics (see Hrouda & Chlupácôvá 1980). K_1, K_2, and K_3 are maximum, intermediate and minimum susceptibilities, respectively.

of magma flow and the short axes perpendicular to it. The orientation of the magnetite grains can then be determined by measuring the AMS.

The directions of AMS vary in distinct Laanila–Ristijärvi and Kautokeino dykes (Fig. 11). In most specimens of dyke LT (Fig. 11a), the maximum axes are almost vertical but the minimum axes have shallow inclinations, thus indicating a vertical direction of magma flow. The Kautokeino dykes (Fig. 11b) show similar behaviour as dyke LT. In Laanila dyke LJ (Fig. 11c) and Ristijärvi dyke RL (Fig. 11d), the minimum axes have steep inclinations and the maximum axes are shallow. Thus, horizontal magma flow may be inferred. However, as Tarling & Hrouda (1993) pointed out, weak anisotropies (*P*) of dykes may cause the maximum and intermediate directions of the AMS ellipsoid to interchange direction and therefore the flow direction is not certain. Many workers (e.g. Ernst 1990; Lister & Kerr 1990; Cadman *et al.* 1993) have shown that the flow direction of initially hot magma of dykes is almost vertical because the liquid magma is less dense than the surrounding solidified country rocks (Lister & Kerr 1990). As the magma cools and gets denser, a horizontal or sub-horizontal magma flow direction may result. It is suggested that the vertical AMS directions of Laanila dyke LT and the Kautokeino dykes could reflect steeper crustal depth than Laanila–Ristijärvi dykes LJ and RL, where the horizontal magma flow direction could be due to emplacement at higher, cooler crustal levels. However, the AMS data is limited and quite scattered and therefore the result is inconclusive.

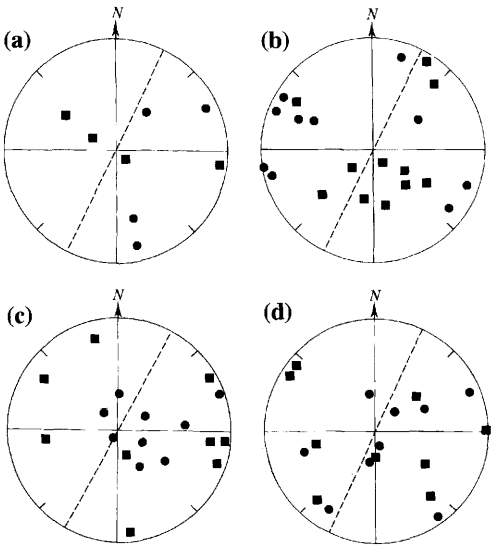


Fig. 11. Directions of minimum (circles) and maximum (squares) susceptibility for specimens of dyke LT (**a**), NA and NB (**b**), LJ (**c**) and RL (**d**), plotted on the lower hemisphere of equal-area stereonets. Average strikes of dykes are given by dashed lines.

Results of high field measurements

In all palaeomagnetic work it is essential that the remanence is stable and thus reliably records the direction of the past geomagnetic field. The stability of the remanence is heavily dependent on the effective magnetic grain size of the remanence-carrying minerals. The highest stability is maintained by small SD grains and the lowest by large MD grains; PSD size grains have magnetic properties between truly SD and MD particles. In the following, the magnetic grain size was determined by measuring the hysteresis properties (Day *et al.* 1977) and by Lowrie–Fuller tests (Lowrie & Fuller 1971; Dunlop 1983). Acquisition of IRM provides information about magnetic grain size characteristics and aids the identification of the dominant magnetic minerals.

The values of saturation remanent magnetization/saturation magnetization (J_{rs}/J_s) and the coercivity of remanence/coercive force (H_{cr}/H_c) of all samples lie in the PSD field (Fig. 12a, Table 7) according to the criteria of Day *et al.* (1977). In empiricial studies (Parry 1980, 1982), it has been shown that mixtures of SD/PSD and MD grain sizes show linear dispersion depending on the proportions of different particle sizes. In the logarithmic plot (Fig. 12b), the hysteresis ratios of the Laanila and Kautokeino samples follow the line of mixtures of the SD and MD magnetite grains (Parry 1982). AF and thermal demagnetization data, which indicated the presence of a large PEF component superimposed on a stable characteristic component, suggested a mixture of grain sizes of SD/PSD and MD type. In Laanila dyke LT, the large PEF component is consistent with the hysteresis properties, thus suggesting an increased proportion of MD magnetite. The smallest magnetic grain sizes (PSD) are found in LJ, LK and KL samples. Samples from these sites also formed the most coherent group of NRM directions and bulk petrophysical properties (Tables 2 and 4). The hysteresis properties of the samples from the Kautokeino dykes are in agreement with those of the Ristijärvi samples. Ristijärvi sample KL shows complex hysteresis properties, probably due to the dominance of weakly magnetic paramagnetic minerals at the expense of ferrimagnetic minerals.

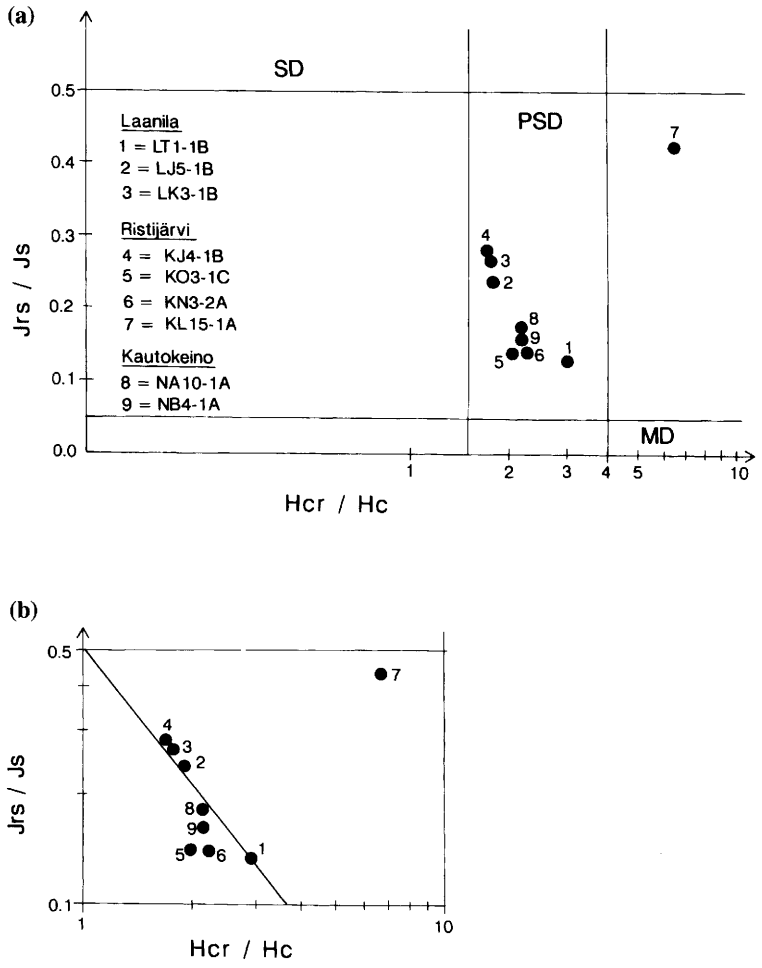


Fig. 12. (**a**) Half-logarithmic and (**b**) logarithmic plots of hysteresis parameters for Laanila–Ristijärvi and Kautokeino dykes. J_{rs} = saturation remanence, J_s = saturation magnetization, H_{cr} = remanent coercivity, H_c = coercive force. Single domain (SD), pseudo-single domain (PSD) and multidomain (MD) fields after Day *et al.* (1977). In (**b**) the line shows the trend for SD and MD mixture, after Parry (1982).

Table 7. *Hysteresis properties of Laanila–Ristijärvi and Kautokeino dykes*

Specimen	J_s(mT)	J_{rs}(mT)	H_c(T)	H_{cr}(T)	J_{rs}/J_s	H_{cr}/H_c
LT1-1B	2.360	0.309	0.008	0.022	0.131	2.960
LL3-1B	12.466	3.344	0.023	0.040	0.268	1.754
LJ5-1B	2.990	0.711	0.025	0.044	0.240	1.790
KN3-2A	11.338	1.585	0.010	0.023	0.140	2.211
KJ4-1B	9.167	2.546	0.024	0.040	0.278	1.697
KO3-1C	6.185	0.876	0.011	0.022	0.142	2.030
KL15-1A	0.083	0.036	0.017	0.109	0.427	6.550
NA10-1A	8.941	1.454	0.013	0.028	0.163	2.174
NB4-1A	8.659	1.546	0.014	0.030	0.179	2.166

NOTE: J_s = saturation magnetization, J_{rs} = saturation remanence, H_c = coercive force, H_{cr} = coercivity of remanence.

The response of SIRM, ARM and NRM to AF demagnetization are compared in the Lowrie–Fuller test. In this test, if the remanence is carried by SD/PSD grains, the normalized weak field ARM is more resistant to AF demagnetization than the SIRM, but has a similar stability to thermoremanence (TRM); MD grains show the opposite behaviour. The NRM of seven specimens from the Laanila–Ristijärvi and Kautokeino dykes was first AF demagnetized in peak fields up to 100 mT. ARM was then produced in a steady field of 0.05 mT superimposed with an alternating field of 100 mT and then progressively AF demagnetized in the same steps as NRM. The specimens were then given a SIRM in the field of 1.4 T (saturation took place below 0.2 T, see Fig. 14) and then AF demagnetized similarly to NRM and ARM.

The NRM of the LJ and LK samples is harder than ARM, which has higher coercivity than SIRM (Fig. 13a and b). In specimens LT (Laanila), KO (Ristijärvi) and NA and NB (Kautokeino), the proportion of VRM component at low coercivities is so high that their NRM intensity decay curves are not easy to compare with the ARM and SIRM curves. However, as in samples LJ and LK, the ARM curves show higher coercivity than the SIRM curves, which is indicative of SD/PSD magnetic grain sizes.

IRM acquisition curves show saturation of all Laanila and Kautokeino samples at 0.1 or 0.2 T (Fig. 14), characteristic of magnetite as the remanence carrier. The rapid increase in IRM in samples KN, KO of Ristijärvi, NA, NB of Kautokeino and especially LT of Laanila shows that the remanence is carried by relatively coarse MD/PSD grains or by a mixture of large MD grains and small SD/PSD grains. Samples LJ, LK and KL show the more gradual increase in IRM, characteristic of a more SD type of remanence. Sample KL is the hardest one and the remanence can be carried by SD magnetite, small MD magnetite or hematite grains (e.g. Dunlop 1981). Thermal demagnetizations showed that magnetite dominates the remanence; thus small PSD magnetite grains are proposed as the remanence carrier for site KL.

Thermomagnetic analysis

The magnetic minerals of the rocks were identified by thermomagnetic analysis (Curie point analysis) by measuring susceptibility versus temperature. Each Curie temperature shown in the thermomagnetic curve characterizes a ferromagnetic mineral in the rock. The Curie temperature can be determined more accurately using the temperature dependence of low-field susceptibility than saturation magnetization, because the susceptibility change at Curie temperature is much more expressive (e.g. Zapletal 1992). Chemical and mineralogical changes can also be observed using the difference in heating and cooling curves. Thus, the suitability of samples for thermal demagnetization (e.g. Schmidt 1993) and for Thellier palaeointensity runs can also be verified.

Samples LL1-1D from Laanila dyke LJ (Fig. 15b) and NA8-1B from Kautokeino dyke NA (Fig. 15c) show a systematic increase in susceptibility and a pronounced Hopkinson peak in the heating curve just before the Curie temperature, as is characteristic of SD/PSD grains. The heating and cooling curves indicate only one ferrimagnetic mineral with a Curie temperature of about 550° C, indicative of Ti-poor magnetite. In sample NB (Fig. 15d) there is also a decline in susceptibility at about 330 °C due to pyrrhotite, which was also identified in thin sections. The heating and cooling curves are

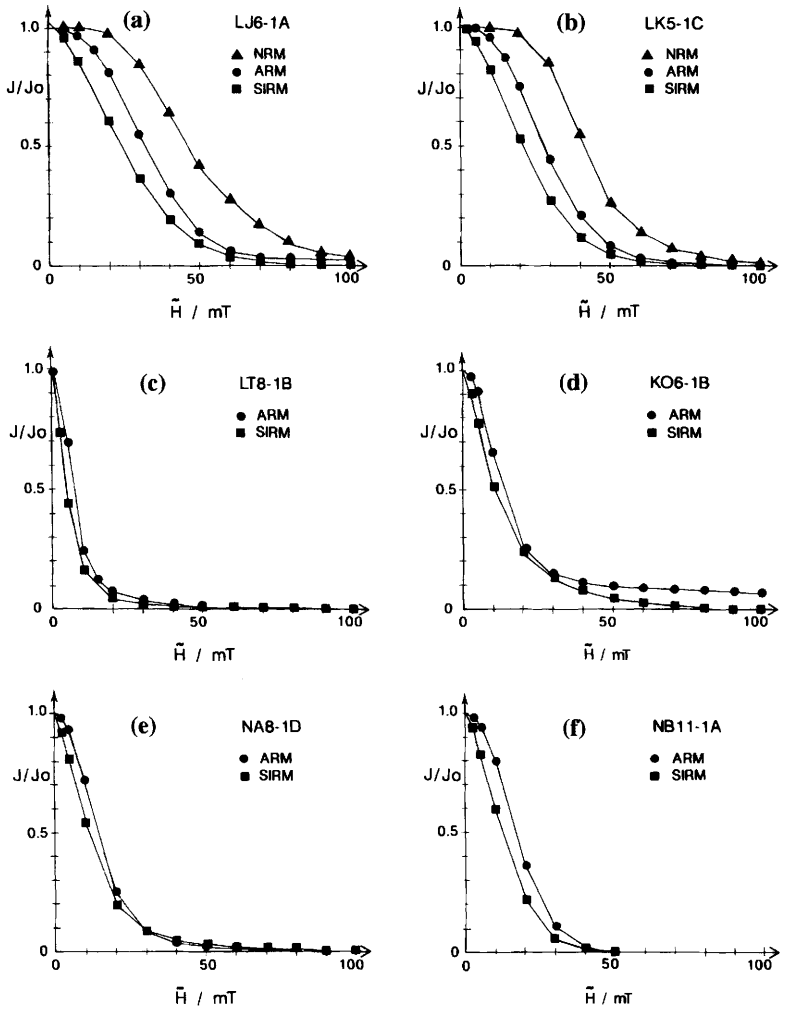


Fig. 13. Lowrie–Fuller test showing NRM, ARM and SIRM curves for specimens of dyke LJ (**a**) and (**b**) and ARM and SIRM curves for specimens of dyke LT (**c**) and RL (**d**) of Laanila–Ristijärvi and NA (**e**) and NB (**f**) of Kautokeino.

irreversible, the cooling curve showing higher susceptibilities. The irreversibility may be due to either changes in internal stresses in the specimen (Zapletal 1992) or the formation of new magnetite (Hrouda 1994) at high temperatures during heating. The difference in the heating and cooling curves is, however, so small that probably no significant chemical changes took place during heating and thus thermal demagnetization of this rock type is suitable.

The thermomagnetic curve for sample LT4-1D (Fig. 15a) is highly irreversible. The Curie temperature determined from the heating curve is about 475°C, indicative of a Ti-rich titanomagnetite, closer in composition to the ulvöspinel end of the magnetite series. In the cooling curve, the susceptibility remains almost constant and is considerably lower than during heating. A slight decrease at about 570 °C may be attributed to the Curie point of Ti-poor magnetite, which formed during heating when titanite or ilmenite were also formed as an alteration product of titanomagnetite.

Petrographic features of LT samples showed that the magnetite phase is almost pure titanomagnetite with no ilmenite lamellae as in dyke LJ. In dyke LT the ilmenite occurs as separate, platy grains. In basic igneous rocks, Ti-rich magnetite is the primary high-temperature mineral (e.g. O'Reilly

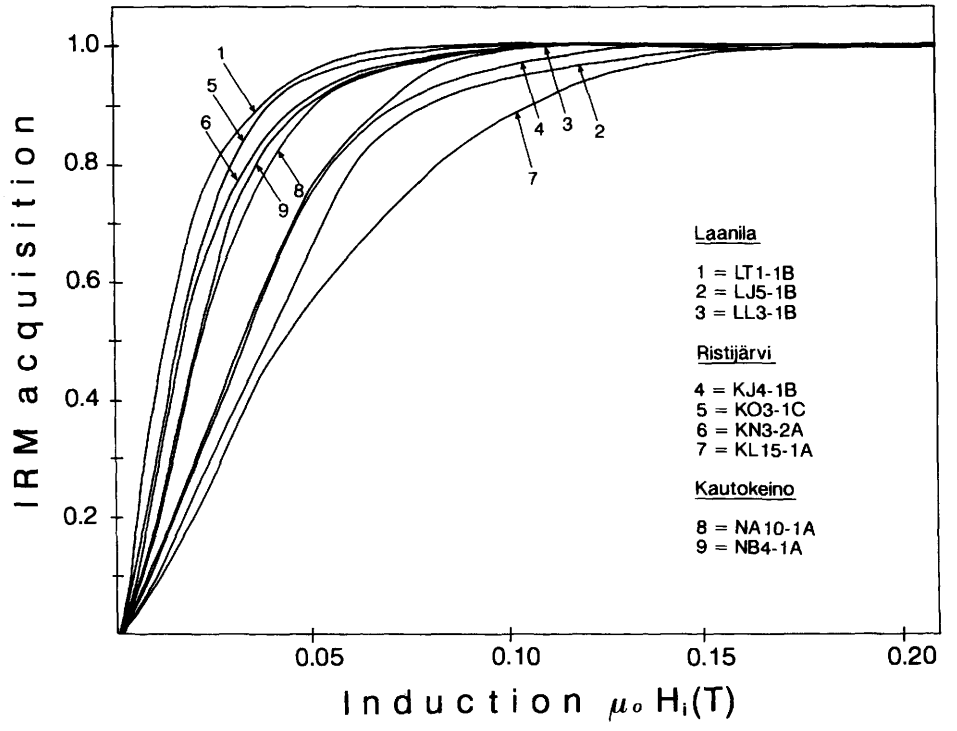


Fig. 14. Normalized IRM acquisition curves for Laanila–Ristijärvi and Kautokeino dykes.

1976). Thus dyke LT represents the initial, most primitive magma of the dyke swarm. The coercive force is known to increase with an increase in the proportion of titanium in titanomagnetite (e.g. Mullins 1977) and therefore susceptibility is also expected to be lower than in pure magnetite. However, the coercive force is lower and the susceptibilities are higher in the LT samples carrying Ti-rich magnetite than in other Laanila samples. The LT samples were dominated by MD magnetite grains (Fig. 12); hence, the grain size dependence of the magnetic properties seems to outweigh the significance of composition. The occurrence of titanomagnetite with no ilmenite lamellae in dyke LT implies that it crystallized in a less oxidizing (O'Reilly 1976) environment than the Laanila–Ristijärvi and Kautokeino dykes.

Results of baked contact tests

The basic requirement for using dyke rocks in palaeomagnetic work is to ensure that the remanence is thermoremanence with a primary ambient field direction acquired during emplacement of the dykes and not during later events. This can be tested by baked contact tests. The baked contact test is based on the idea that when a host rock at the contact of an igneous intrusion is heated above the blocking temperatures of the minerals carrying the remanence, they will acquire a new remanence in the direction of the igneous intrusion. The maximum temperature reached in a host rock close to an igneous intrusion as well as the stability of the baking magnetization decrease with distance from the contact (Everitt & Clegg 1962; McClelland-Brown 1981). As Schmidt (1991) pointed out, palaeomagnetic contact tests demand ideal behaviour of both the dyke and the host rock. The Laanila–Ristijärvi and Kautokeino dykes do behave ideally but, as often happens in Archaean felsic gneiss areas, the stability of the NRM in the host rocks is weak. Therefore, a full baked contact test comprising magnetically stable contact, hybrid and unbaked zones at the sampling sites could not be carried out. However, by compiling known remanence direction data on the host rock from elsewhere in the granulite belt and from the Archaean basement area (e.g. Pesonen *et al.* 1989) and by combining it with petrophysical information, we can evaluate the primary nature of the remanence in the dykes.

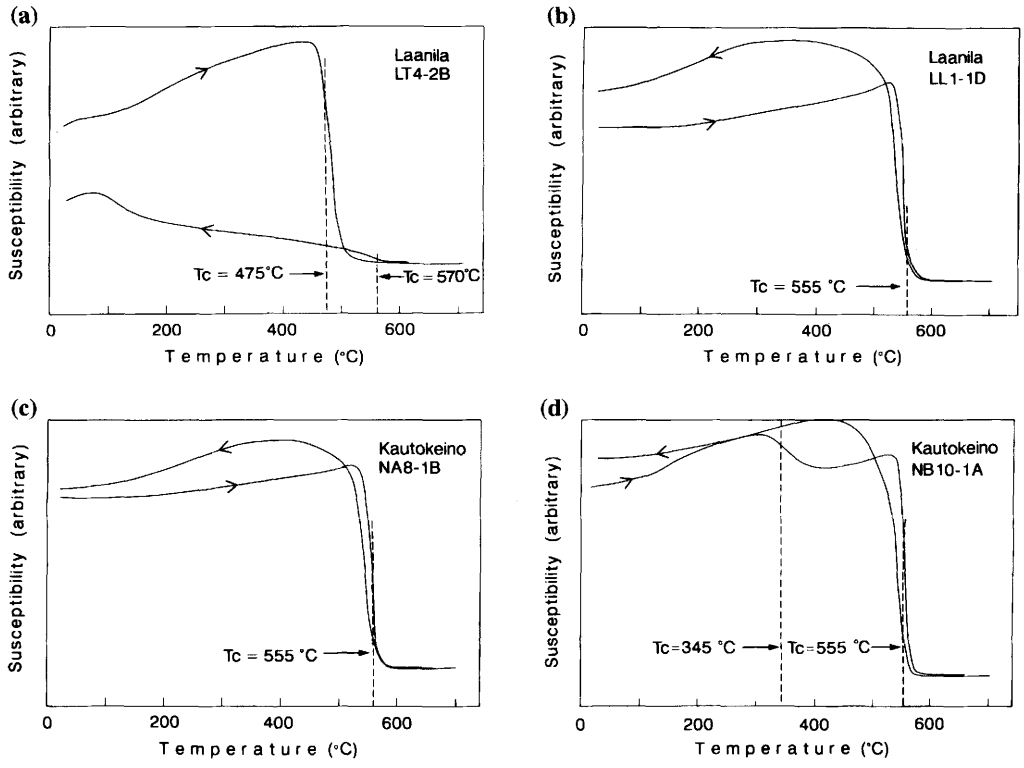


Fig. 15. Thermomagnetic curves (susceptibility/ temperature) for specimens of Laanila–Ristijärvi and Kautokeino dykes. T_c denotes Curie temperature. Specimen LT from dyke LT contains Ti-rich titanomagnetite as main magnetic mineral. All the other dykes contain Ti-poor titanomagnetite.

Laanila–Ristijärvi dykes

Baked contact tests were carried out for all dykes. At site LJ, the remanence direction of the granulite right at the contact (Fig. 16a) is similar to that in the diabase dyke (Table 2) indicating that the remanences are coeval. Further from the contact (Fig. 16b), at the hybrid zone, the baking effect is still seen as negative inclination, although the directions are highly scattered. Probably due to the scattered data and the instability of the host rock, the high temperature component representing the host direction and the low temperature component representing the dyke direction (pTRM) could not be isolated. At site LJ, the baking effect is also seen as increased density, remanence intensity and susceptibility values in the contact, whereas the unbaked sample shows clearly lower values (Table 4, Fig. 8).

At site LT the granulite at the contact exhibits a direction close to that of the dyke, although the magnetization is poorly defined (Fig. 17). Further from the contact no stable directions were obtained. At this site, the density, remanence intensity and susceptibility are slightly lower in the baked sample than in the unbaked sample. The reason for the contrasting baking effects at sites LJ and LT may be the temperature difference between the dyke and host rocks at these sites. At site LJ, the dyke was evidently emplaced in a cooler environment and thermally affected the host rock. At site LT, where the host rock temperature was probably higher, the baking effect of the dyke was relatively low.

At site LT, where the unbaked host granulite was sampled, the remanences were too weak to give reliable results. However, the remanence direction of quartz diorite at Akujärvi (U–Pb zircon age 1925 Ma, age given in Meriläinen 1976; Pesonen & Neuvonen 1981) in the granulite belt, about 15 km from the Laanila dyke, points moderately downwards towards the northwest and thus clearly deviates from the Laanila dyke and baked contact directions. Further west, in the granulite of Menesjärvi (age *c.* 1850 Ma, Meriläinen 1976; Papunen *et al.* 1977; Pesonen *et al.* 1989), about 80 km from Laanila, the remanence direction corresponds to that at Akujärvi.

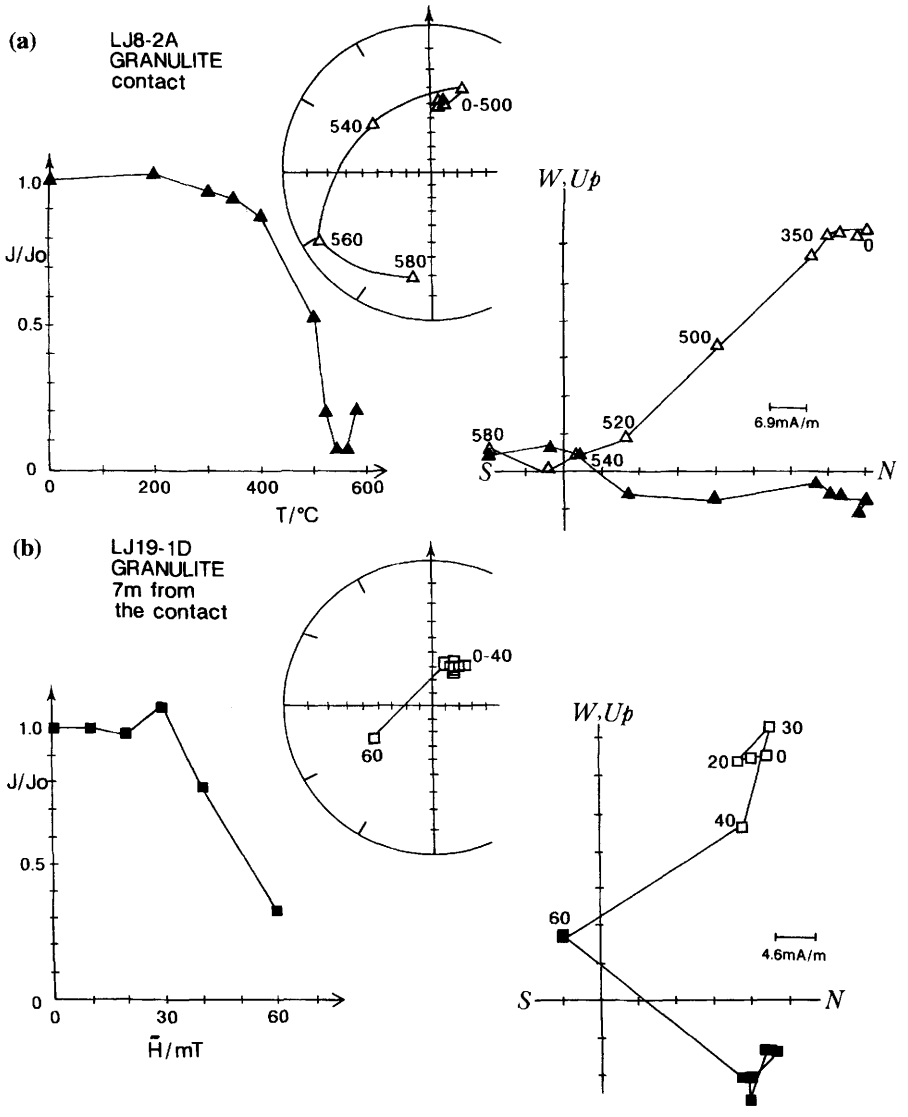


Fig. 16. Baked contact test for dyke LJ (width 20 m) in Laanila. Host granulite right at contact (**a**) carries similar magnetization to that of diabase dyke. Baking effect is still seen *c.* 7 m from the contact (**b**).

Contact samples were taken at sites KL, KM and KP from Ristijärvi dyke RL. At site KL, the remanence direction of the Archaean gneiss accords with the dyke direction (Table 2) and, hence, at this site, too, the remanences were acquired at the same time. The remanence direction of amphibolite at site KM is close to the mean Ristijärvi dyke direction (Table 2) but the inclination is much shallower and the declination points more to the west. In the amphibolite at site KP, the remanence directions are scattered, and reliable results could not be obtained. At Ristijärvi, samples were not collected far from the dyke and therefore a full baked contact test could not be carried out.

The results of the baked contact tests combined with the results from the unbaked Akujärvi and Menesjärvi rocks indicate that the ChRM is of thermal origin (TRM) in the Laanila–Ristijärvi dykes and baked contacts. Further support for TRM comes from a Thellier palaeointensity determination on one Laanila specimen, which

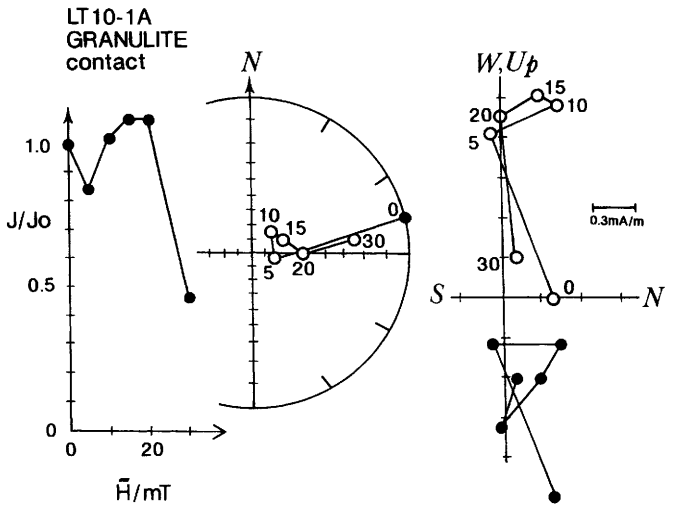


Fig. 17. Baked contact sample from granulite host rock of dyke LT.

yielded a reliable Arai plot with a palaeointensity of 29.6 ± 0.9 μT, consistent with global palaeointensity data about 1.0 Ga ago.

Kautokeino dykes

The Kautokeino dykes have baked the host rocks. This is especially clear in the contact amphibolite of site NA where the remanence direction agrees with the dyke remanence direction (Fig. 18). However, the direction is still close to the dyke direction in a sample taken 17 m from the contact. This can be interpreted in at least two ways. According to baked contact models (e.g. Schwarz 1977), the host rock should acquire partial thermoremanence (pTRM) in an area of about half the dyke width. Therefore, the baking effect of the dyke has extended as far as 17 m from the contact because the width of the dyke is at least 34 m rather than the 22 m consumed initially. Alternatively, a hidden dyke belonging to the same dyke swarm as dyke NA has baked the host rock. The Kautokeino dykes seem to occur in a comparatively narrow zone, but it is possible that the dyke swarm includes a number of dykes that are not exposed and are not seen on the aeromagnetic maps because of the high proportion of VRM aligned in the PEF direction. Stable remanence directions were not obtained from the host amphibolite and therefore, at this site baked contact test is inconclusive. At site NB where the host rock is Archaean gneiss, the remanence direction of the unbaked gneiss differs clearly from the direction of the dyke (Table 3) and thus, baked contact test is positive.

Discussion

Interdyke variation within the Laanila swarm

The easternmost of the Laanila dykes (LT) differs from the westernmost dyke (LJ) on palaeomagnetic, petrographic and bulk petrophysical criteria. Secular variation may explain some of the difference in remanence directions (Fig. 5) but does not explain the other observed differences. The differences in petrography and petrophysics may be due to the fact that dykes LT and LJ represent different crustal levels.

Petrographic and rock magnetic features indicate that the magnetite grains in dyke LT are coarser than those in LJ suggesting that LT crystallized at deeper levels than dyke LJ. Fluid activity is more pronounced at higher crustal levels than at lower crustal levels. This is seen in the Laanila dykes, where LJ (and RL of Ristijärvi) reveals more pronounced hydrothermal alteration (seen as thorough sericitation of plagioclase) than LT. On the basis of AMS data, a vertical magma flow direction for dyke LT and a horizontal direction for dyke LJ can be inferred (although not conclusively). Baked contact tests at dykes LT and LJ show that at the time of dyke injection, the host rock temperature must have been lower than the dyke temperature, that is, below the Curie point of magnetite (*c.* 580 °C). Furthermore, the petrophysical features of the host rocks imply that at site LT the host rock may have been hotter than at site LJ. Low altitude aeromagnetic maps reveal a strikingly clear difference between the magnetic field strengths west and east of dyke LJ. According

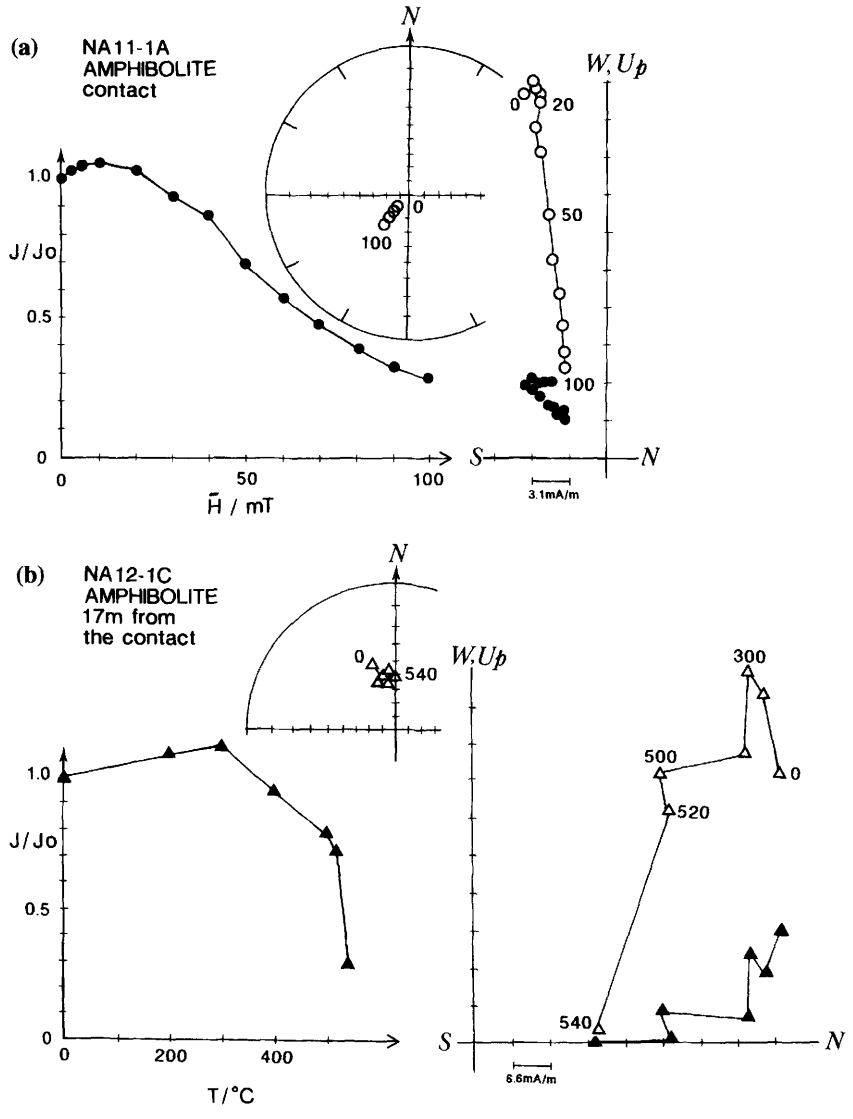


Fig. 18. Baked contact test for dyke NA (width at least 22 m) in Kautokeino. Host amphibolite right at contact (**a**) carries similar magnetization to that of diabase dyke. Similar remanence direction was isolated *c.* 17 m from contact (**b**).

to magnetic modelling by Airo (1994), this dyke represents a deep fracture between and separating two blocks at different crustal levels.

From the foregoing indications we infer that the remanent magnetization blocking temperature was reached in the LJ dyke prior to have reached in the LT dyke. Since the dykes were emplaced, dyke LT has undergone a greater degree of exhumation than dyke LJ, and thus a deeper crustal level of this intrusion is now exposed. Halls & Palmer (1990) have described a similar case from the 2.45 Ga old Matachewan dykes in Canada, where the N polarity dykes have been relatively uplifted along faults and represent a deeper expression of the swarm than the R polarity dykes. The importance of the finding at Laanila is that the crustal block containing dyke LT must have been uplifted **after** the Laanila dyke was emplaced, that is, after about

1.0 Ga. On the APWP (Fig. 19) the pole LT is slightly younger than poles LJ and RL; this may be accidental but is nonetheless consistent with the above model.

Palaeomagnetic poles

The Laanila–Ristijärvi, Kautokeino and Karasjok dykes trend NE–SW and display a negative anomaly on aeromagnetic maps. Hence, there is a possibility that all the dykes belong to the same diabase dyke swarm with a width of at least 200 km. Sm–Nd ages ranging in 1066–1013 Ma are coeval within the error limits. Bulk petrophysical properties, petrography, thermomagnetic and high field analysis also show a similarity between the Laanila and Kautokeino dykes. Furthermore, preliminary geochemical results show that the Kautokeino and Laanila–Ristijärvi dykes represent same tholeiitic basalts from initial rifting regimes in continental settings (P. Pihlaja, pers. comm., 1995). However, the remanence directions and pole positions of the dyke swarms differ significantly (Figs 5 and 19), suggesting a clearly different magnetization age for the Laanila–Ristijärvi and Kautokeino dykes. From the scatter of virtual geomagnetic poles (VGPs) of three Laanila dykes and from two Kautokeino dykes, it can be considered that the secular variation has been smoothed out for both dyke swarms.

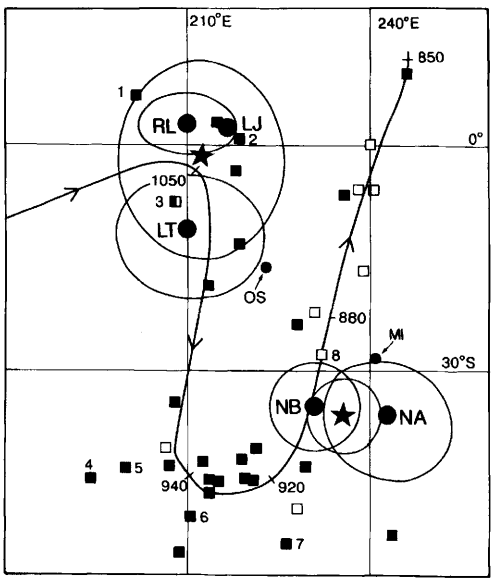


Fig. 19. Apparent Polar Wander Path (APWP) for Fennoscandia (1050–850 Ma) based on data from Elming *et al.* (1994). Closed (open) symbols denote normal (reversed) polarity; half open symbols denote mixed polarity. Stars represent mean poles for Laanila–Ristijärvi and Kautokeino dykes with their A95 confidence circles. Circles represent dyke mean virtual geomagnetic poles (VGPs) shown with their confidence ovals. Poles OS and MI are calculated from uncleaned NRM data from Kautokeino dykes (Olesen & Sandstad 1993) and Karasjok dykes (Midtun 1988), respectively. Squares denote palaeomagnetic poles from Sveconorwegian rocks of Fennoscandia (Pesonen *et al.* 1991; Elming *et al.* 1994). Numbered poles are referred to in text.

The latest APWP of Fennoscandia (Elming *et al.* 1993) is based on well-defined palaeopoles. However, due to variety of used dating methods, which in many cases are other than the precise U–Pb method, the ages of the poles are still controversial. Therefore, the APWP cannot be used to give accurate magnetization ages of the poles although their relative ages can be inferred.

The Laanila–Ristijärvi and Kautokeino poles plot among the poles obtained from the Sveconorwegian Southwest Scandinavian Domain in southern Sweden and Norway (see e.g. Poorter 1975; Bylund 1981,1985,1992; Stearn & Piper 1984). Some of the poles derive from Sveconorwegian intrusions, and some record the uplift and cooling of the Sveconorwegian terrain (see e.g. Pesonen *et al.* 1991). Thus far, the palaeopoles from the Laanila and Kautokeino dykes are the only poles in the Fennoscandian palaeomagnetic database (Pesonen *et al.* 1991; Elming *et al.* 1993, 1996) outside and not close to the Sveconorwegian area in Fennoscandia.

Neoproterozoic dyke magmatism in Fennoscandia coincides with the Sweconorwegian–Grenvillian Orogeny 1250–850 Ma ago. The onset of the orogeny occurred in southern–central Fennoscandia and is represented by the Central Scandinavian Dolerite Group in Sweden and Finland with ages of 1270–1200 Ma (Gorbatschev *et al.* 1987). Dyke complexes of early stages also occur in Lake Ladoga and west of Lake Onega in Russia, in northern Sweden and in southern Norway, but they have not been dated definitely (Gorbatschev *et al.* 1987).

In the later stages of the Sveconorwegian Orogeny (1050–850 Ma), several dyke swarms intruded in the Fennoscandian shield. Dyke injecton was most extensive in southern Fennoscandia about 1050–850 Ma ago (Rb–Sr and Sm–Nd ages; Patchett & Bylund 1977; Johansson & Johansson 1990). It is represented by the Blekinge–Dalarna dolerite dykes (BDD), which are aligned parallel to the Protogine Zone between the Sveconorwegian and Svecofennian parts of the Fennoscandian shield. Some of the BDD dykes have been dated and their palaeopoles determined within the Protogine Zone, west and east of the zone and also in the Svecofennian host rocks (e.g.

Bylund 1981, 1992; Pesonen *et al.* 1991; Bylund & Elming 1992). The Laanila and Kautokeino dyke swarms belong to the same age group as the BDD dykes.

The oldest palaeopoles from the Sveconorwegian Southwest Scandinavian Domain derive from the Bamble intrusions, southern Norway, which comprise basic igneous rocks, amphibolites and hyperites (Poorter 1975; Stearn & Piper 1984). K–Ar dates of the intrusions are in the range 1080–950 Ma. Rb–Sr whole-rock dates give an age of 1139 ± 100 Ma. In the palaeomagnetic grading system (Briden & Duff 1981), three palaeopoles (poles 1–3, Fig. 19) from the Bamble intrusions are regarded well defined but considering the dating methods, the ages of the palaeopoles probably do not record the intrusion ages, but are younger. The poles correspond to the ones from the Laanila–Ristijärvi dykes.

The Sveconorwegian APWP makes a steep loop at 950–900 Ma and turns to lower latitudes from about 920 Ma onwards (Fig. 19). The pole from the Kautokeino dykes is situated on the younger part of this loop. The only poles with precise U–Pb zircon ages are located at the southernmost apex of the loop and they all derive from the Rogaland igneous complex, west of the Protogine Zone in southern Norway. Pole 4 (Fig. 19) derives from the Egersund farsundite and yields U–Pb ages on zircon of 977 ± 36 Ma and 944 ± 84 Ma (Murthy & Deutsch 1975). A new U–Pb age of zircon gives 928 ± 2 Ma for the Egersund anorthosite (Duchesne *et al.* 1993). Pole 5 has been obtained from the Åna-Sira massif yielding a U–Pb age of zircon of 930 ± 4 Ma (Duchesne *et al.* 1993). Pole 6 derives from the Bjerkreim–Sokndal lopolith with a U–Pb age on zircon of 945 ± 40 Ma (Stearn & Piper 1984). Pole 7 represents a mean pole from the Rogaland farsundite with U–Pb age of zircon of 944–977 Ma (Stearn & Piper 1984). A palaeopole at the lower latitudes (pole 8, Fig. 19), near the Kautokeino pole, is from the Egersund olivine dolerite dyke of southern Norway (Storetvedt 1966*a*,*b*). Palaeomagnetic data suggest a Sveconorwegian age for these dykes (see e.g. Pesonen *et al.* 1991), but a Rb–Sr determination provides an age of 630 Ma (Sundvoll 1987).

Hence, all the palaeopoles at high southern latitudes of the loop yield isotope ages younger than 1.0 Ga except the new pole from Kautokeino dykes which have an exceptionally high Sm–Nd age of 1066 ± 34 Ma. Regarding the markedly different pole position of the Kautokeino dykes compared to the pole of the 1042–1013 Ma old Laanila–Ristijärvi dykes and the above presented precisely dated poles, it is evident that the Kautokeino dykes yield a younger magnetization age than the Laanila–Ristijärvi dykes. On the basis of the dated poles, the age of the Kautokeino pole is *c.* 900 Ma. In order to explain the deviating pole position of the Kautokeino dykes with respect to the Laanila–Ristijärvi dykes, three scenarios are proposed.

First, the Kautokeino palaeopole may record later local tectonic movements which have taken place after the dyke emplacement resulting in new remanence directions. The pole positions are thus fortuitous. Olesen & Sandstad (1993) report large-scale Proterozoic fractures and Phanerozoic postglacial fault zones within the Kautokeino greenstone belt which may be responsible for tilting of the dykes. The Kautokeino dykes dip 60–70° southeast, but tilting to vertical position would steepen the inclination and not make it shallower as required. Furthermore, since the remanence directions from the similarily striking Kautokeino and Karasjok dykes are similar, it is likely that they have been emplaced in their present position. Hence, the present dips of the dykes do not explain the deviations.

Another reason for the younger magnetization age may be due to delayed blocking of the remanence of the Kautokeino dykes. By analogy with the scenario for the Laanila dykes LT and LJ (see before), it is possible that at the present erosion surface the Kautokeino–Karasjok dykes represent a much deeper level than the Laanila–Ristijärvi dykes. At deeper crustal level the magnetization would have been delayed due to prolonged cooling of the dyke at higher ambient temperatures. However, considering the magnetization ages of about 1030 Ma and 900 Ma of the Laanila–Ristijärvi and Kautokeino dykes, respectively (Fig. 19), this model requires blocking of the remanence more than 100 Ma later at deeper parts of the dyke than at upper parts, which is not possible. Moreover, if the difference of crustal levels is due to late vertical uplift of the Kautokeino–Karasjok block, occurrence of steep vertical faults between the Laanila and Karasjok areas as well as differences in the magnetization and gravity anomalies is expected. However, there are no clear indications of such faults and differences.

Third, the isotope ages of the Laanila–Ristijärvi and Kautokeino dykes are consistent with the palaeomagnetic ages but the previous APWP of Fennoscandia (Fig. 19) is wrong and should be revised. This would require a self-closing loop and a much more complex APWP than the previous one.

With the present data, none of the models is favoured. Further isotope age determinations with U–Pb (zircon or baddeleyite) or ^{39}Ar–^{40}Ar techniques and palaeomagnetic measurements of both Kautokeino and Laanila–Ristijärvi dykes are required to test the credits of these models.

Conclusions

(1) Baked contact tests of the Laanila–Ristijärvi and Kautokeino dykes show that the characteristic remanence of the dykes is primary TRM acquired during cooling of the magma. TRM is supported by similarity of the ARM with NRM, and enhanced Q values of the baked contact rocks.

(2) The petrophysical, petrographic and magnetic properties of the Kautokeino dykes show similarities with the Laanila dykes and thus both dyke sets may originate from the same magma source. Similar remanences also occur in the dykes of Karasjok, between Laanila and Kautokeino, indicating the vast extent of Neoproterozoic diabase dykes in the northern Fennoscandian shield.

(3) The magnetic and petrographic properties of Laanila dyke LT differ from those of Laanila dyke LJ and Ristijärvi dyke RL. The differences are interpreted as indicating that, at the present erosion surface, dyke LT represents a deeper horizontal level and yields a younger magnetization age than dykes LJ and LK. The granulite block containing dyke LT has been uplifted since the Laanila dykes were emplaced.

(4) Sm–Nd isotope analyses provide ages of 1042 ± 50 Ma, 1013 ± 32 Ma and 1066 ± 34 Ma for the Laanila, Ristijärvi and Kautokeino dykes, respectively. Magnetization ages, suggested from the Fennoscandian APWP show that the Laanila dykes are about 100 Ma older than the Kautokeino dykes. More investigations are needed in order to explain the discrepancy between the palaeomagnetic and isotope ages of the Kautokeino dykes.

Tapani Mutanen and Pekka Pihlaja chose the sampling sites at Laanila and Ristijärvi and carried out some of the petrographic studies. Odleiv Olesen provided sample material from Kautokeino and Karasjok and acted as guide in the field. Kalle Neuvonen provided preliminary data on the Laanila dykes. Meri-Liisa Airo was consulted about the aeromagnetic maps. Matti Leino assisted in many ways in the palaeomagnetic laboratory and Hilkka Kumpunen made the palaeomagnetic measurements. Salme Nässling drafted the figures. The language was corrected by Gillian Häkli. Andrew Cadman and Sten-Åke Elming made valuable comments on the manuscript. To all these people we express our sincere gratitude.

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